



GLOBALFOUNDRIES 22FDX Standard Cells 8T

User Guide

DesignWare® Logic Libraries
IVCH-SC-8T-G22FDX-A1

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Revision History



Note

Links and references to section, table, figure, and page numbers in this table are only assured to be valid for the version in which the change is made.

Date	Document Version	Description
November 2018	0.50	Initial Release
November 2018	0.60	Added Custom Mixed VT Synchronizers
April 2019	0.65	Updated the following sections: ■ Library Architecture ■ Usage Scenario of Level Shifter Cells ■ Observations Updated the following tables: ■ Table 2-1 Electrical Specifications ■ Table 2-2 Electro-Migration Specifications ■ Table 3-3 Electrical Characteristics of Header Style Power Switches - SLVT ■ Table 3-4 Electrical Characteristics of Header Style Power Switches - RVT ■ Table 3-5 Electrical Characteristics of Mother-Daughter Header Style Power Switches - SLVT ■ Table 3-6: Electrical Characteristics of Mother-Daughter Header Style Power Switches - RVT ■ Table 4-2 Simulation Specifications and Criteria ■ Table 4-3 Maximum Supported Frequency ■ Table 6-2 Maximum Transition Supported for RVT/HVT Non-Shift Products
August 2019	0.70	Updated Power Shut Off with State Retention on page 25 section Updated the following tables: ■ Table 3-2 LPK and Shift Cell Set ■ Table 4-1 Summary of Supported Cells and Intended SoC Level Usage
November 2020	0.80	Updated ECO Kit information and document template.

Date	Document Version	Description
November 2021	0.90	Updated section “Power Shut Off with State Retention”. Applied template changes to conform to Synopsys standards.
April 2022	1.00	Updated information on Retention flops.

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General Description

1.1 Introduction

This document describes the Synopsys Standard Cells library in GF22FDX technology for 8-Track platform. The platform and their supported products, cell types, and use cases at design level are described in detail in the subsequent sections.

1.2 Library Architecture

[Table 1-1](#) describes the Standard Cell architecture with key parameters like:

- Height
- Width
- Poly Pitch
- Layout Style for RX

The following figure shows the 8T Cell Architecture.

Figure 1-1 8T Cell Architecture

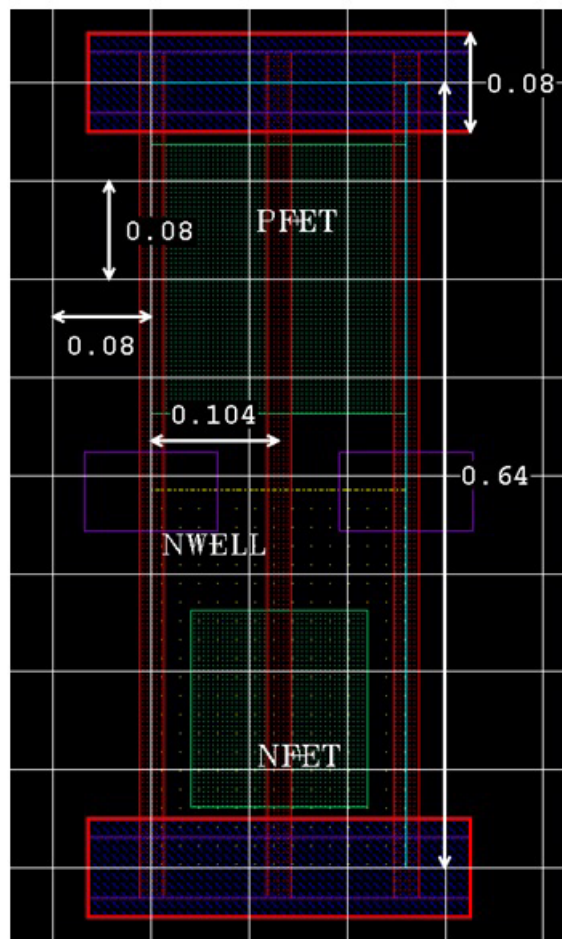


Table 1-1 12-Track Architecture Parameters

Offer Type	Track	V _T Support	Channel Length Support	Layout Style	Height	Poly Pitch(nm)	Min Device Width	PG Rail	Well Orientation
Density	8 T	SLVT LVT	20 nm, 24 nm, 28 nm,	PMOS - Continuous RX NMOS - SDB	640 nm	104 nm	100 nm	M1 80 nm	Flip-Well ¹
		RVT HVT							Regular-Well ²



Note

¹Flip-Well = PMOS in PWell, NMOS in NWell

²Regular-Well = PMOS in NWell, NMOS in PWell

1.3 Product Overview

The following table provides the Product classification for 8T library.

Table 1-2 Product Overview

Product Offer	Description
Base Kit (BASE)	Combinational and Sequential cells, along with physical cells.
Low Power Kit (LPK)	Cells required to enable various low power flow implementation methodologies.
Bi-directional Level Shifter Kit (Shift)	Bi-directional voltage level shifting cells (Capable of shifting both High-to-Low as well as Low-to-High).
High Performance Kit (HPK)	Differentiated cells to be used along with Base library to improve design level PPA.
Engineering Change Order Kit (ECO)	Cells required for metal ECOs, to realize various logic functions using only few metal mask updates.

1.4 Library and Logic Variants

- Library Variants
- Cell Count
- Functionality Richness

Table 1-3 8-Track Library and Logic Variants

Offer Type	Track	V _T Support	Library Variants	PG Rail	Well Orientation
Density	8T	SLVT LVT	Base Kit High Performance Kit (HPK)	1300+ Cell per VT/Length Functionalities ~190	Combinational: ~120 Sequential: ~32 PnR Enablement: ~20 LPFlow Enablement: ~30
		RVT HVT	Low Power Flow Kit (LPK) Level Shifter (Shift and ShiftHL) Engineering Change Order Kit (ECO)		

1.5 Logic Portfolio

- Combinational Cells
- Sequential Cells
- Low Power Flow Enablement Cells
- ECO Cells
- P and R Enablement Cells

Table 1-4 Logic Portfolio

Sequential Cells	
Non- Scan Latch	
Non- Scan Flip-flop	Set
	Reset
	Set- Reset
Scan Flip-flop	Set
	Reset
	Set- Reset
Clock Gates	Single CLK
Multi-Bit Flip-flop (2-bit and 4-bit)	Set
	Reset
Clock Synchronizers (2, 3 and 4 Stage)	Set
	Reset
Low Power Flow Enablement Cells	
Power Switch	Header
	Footer
Always-On (AON)	Buffer
	Inverter
	Latch
Isolation Cells	Clamp 0/1
	Clamp 0/1

Table 1-4 Logic Portfolio

Level Shifters	Source placed
	Destination placed
	Isolation
Retention Flip-flop	
ECO	
Metal fix ECO Cells	
P and R Enablement	
Antenna	Decap
Tap	Filler
Row end	Column end

1.6 Model Support

- Timing/ Power/ Noise/ Variation Models
- Logic, DFT Models
- Abstract, Physical Models

The following table shows the supported models and design flows.

Table 1-5 Supported Models

Views	Support
Verilog	√
Power-Aware Verilog	√
NLDM/NLPM	√
CCS Timing, Power, Noise	√
AOCV	√
LVF	√
PGV (Cadence-Voltus)	√
APL (Ansys-RedHawk)	√
Milkyway, NDM	√
Fastscan, Tetramax	√

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Electrical Specifications

The following are the electrical specifications supported of 8T.

- VT
- Channel Length Options
- Voltage
- Temperature Range
- Body Biasing

Table 2-1 Electrical Specifications

Offer Type	Track	VT Support	Voltage Range Support	Temperature Range Support	Body Bias Support	Channel Length Support	Devices
Density	8 T	SLVT/LVT	Overdrive: 0.90 V	-40 °C to 125 °C	Static FBB (Up to 0.90 V)	20 nm, 24 nm, 28 nm	slvtpfet, slvtnfet
			Nominal: 0.80 V				lvtpfet, lvtnfet
			Under drive: 0.65 V				pfet, nfet
		RVT/HVT	Overdrive: 0.90 V		-		hvt pfet, hvtnfet
			Nominal: 0.80 V				

2.1 Electro-Migration Specifications

The following table provides electro-migration specifications

Table 2-2 Electro-Migration Specifications

Parameters	Values	
	LVT/SLVT	RVT/HVT
Process	Fast (FFG)	Fast (FFG)
Voltage	0.99 V	0.99 V
Body Bias (VNW, VPW)	0 V, 0 V	0 V, 0 V
Temperature	100 °C	100 °C
VT Type	SLVT	RVT
Channel Length	20 nm	20 nm
Frequency	CLK: 1.80 GHz	CLK: 1.20 GHz
	DATA: 0.60 GHz	DATA: 0.40 GHz
Power-On Hours	100 KPOH	100 KPOH
Load	CLK: FO15	CLK: FO15
	DATA: FO25	DATA: FO25

3

Low Power Kit (LPK)

3.1 Low Power Techniques and Supported Cells

The following table lists the low-power techniques and cells provided by LPK for various low-power design implementation techniques.

Table 3-1 Low Power Techniques and Supported Cells

Low Power Technique	Standard Cell IPs Provided	Description
Multiple supply voltages	Level Shifters	Each block operating at different voltages, still meeting the overall SoC performance. Reduces the dynamic and leakage power consumption in the blocks.
Power shut-off Power gating	Power Switch, Isolation, and Always-On cells	Power gating for few blocks with isolation to other Always-On blocks.
Power shut-off with state retention	Retention Flops	Power shut-off with retaining previous state for faster wake up.
Power shut-off with state retention and multiple supply voltages	Isolation level shifters	Level shifters with Isolation along with cell offered in Power shut-off with state retention.

Table 3-2 LPK and Shift Cell Set

Cell Name	Category
AONBUF	Always-On Cell
AONINV	
AONLATQ	

Table 3-2 LPK and Shift Cell Set (Continued)

Cell Name	Category
AONISOHIH	Isolation Cell
AONISOLOH	
ONISOHI	
OSOISOLO	
LSS	Bi-Directional level shifter
LSD	
LSSCL0	Bi-Directional Isolation level shifter
LSSCL1	
LSDCL0	
LSDCL1	
LSSHL	High-to-Low level shifter
LSDHL	
LSSHLCL0	High-to-Low Isolation level shifter
LSSHLCL1	
LSDHLCL0	
LSDHLCL1	
RET1SDFFQ	Retention Flip-flop
RET1SDFFRQ	
RET1SDFFSQ	
RET2SDFFQ	
RET2SDFFRQ	
RET2SDFFSQ	
HEADER	Power Switch
HEADERBUF	
FOOTER	
FOOTERBUF	
HEADERMD	Power Switch with Mother-Daughter Architecture

3.2 Multiple Supply Voltage (MSV)

Level shifters are used to shift input signals from one voltage domain to another. This may be from “low to high”, or “high to low”.

The type of level shifter cells available in the current offering include bi-directional ones that are capable of acting as high-to-low or low-to-high, and cells that can only perform high-to-low level shifting.

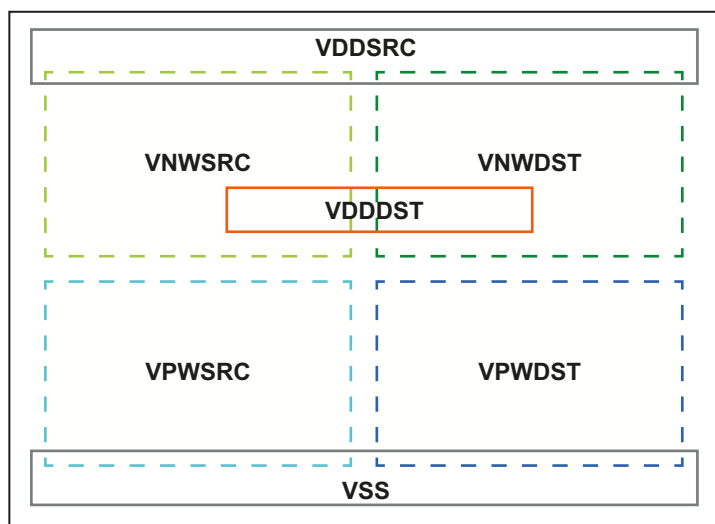
Other categorizations of the level shifter cells based on the power rail connections include:

- Source-side placed level shifter
- Destination-side placed level shifter

3.2.1 Source-Side Placed Level Shifter

As shown in the following figure, the main power supply rail in the Source-side placed level shifter is the source domain, while the destination power rail is internal to the cell.

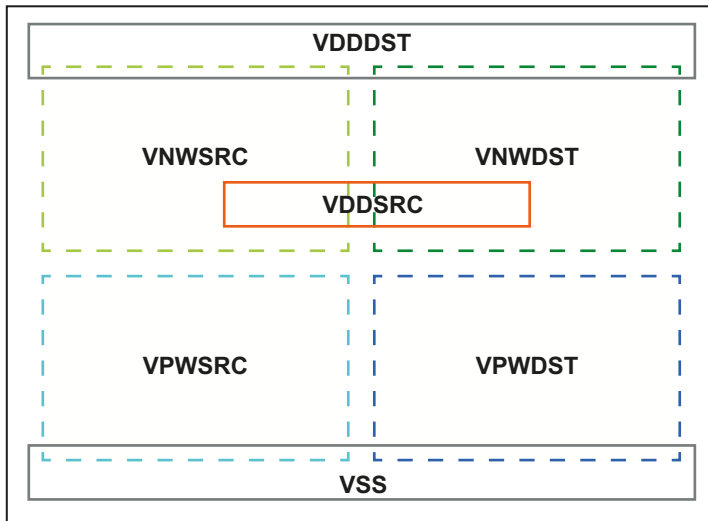
Figure 3-1 Source-Side Placed Level Shifter



3.2.2 Destination-Side Placed Level Shifter

As shown in the following figure, the main power supply rail in the Destination-side placed level shifters is the destination domain, while the source power rail is internal to the cell.

Figure 3-2 Destination-Side Placed Level Shifter



Well Connection Strategy

- VNWSRC/VNWDST should be connected to N-well bias voltage placed in domain
- VPWSRC/VPWDST should be connected to P-well bias voltage placed in domain
- In GF22FDX technology, default N-Well and P-Well voltage level is 0 V

3.2.3 Level Shifter Cells

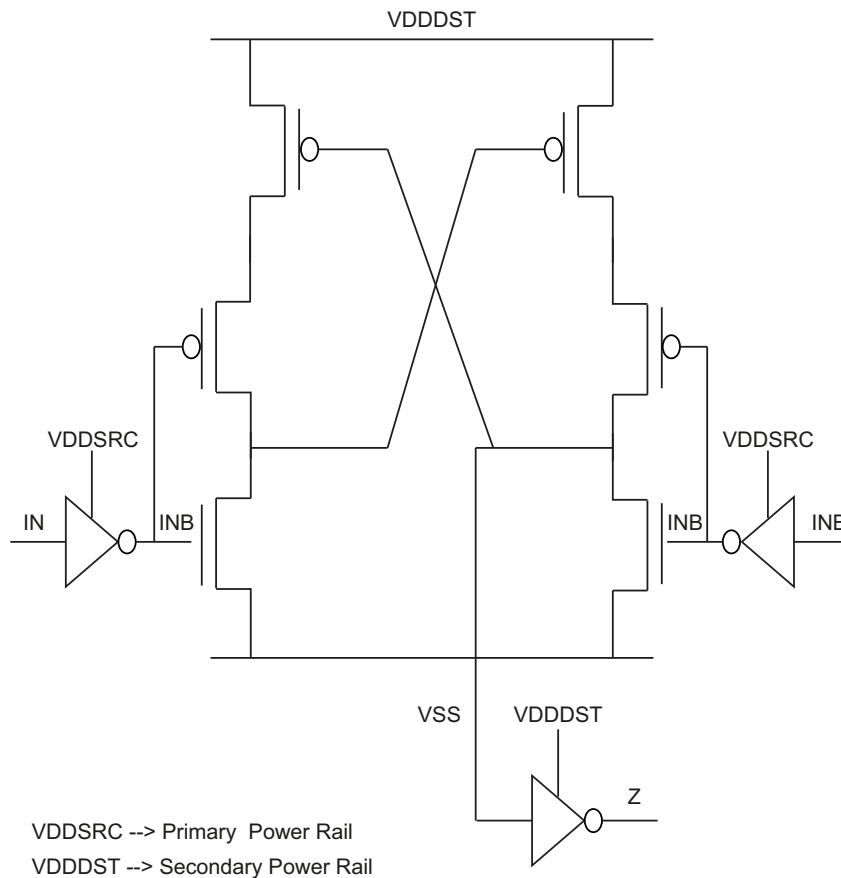
3.2.3.1 LSS Cell

Function

It is a bi-directional voltage level shifter capable of shifting from low-to-high as well as high-to-low. It has to be placed in source domain.

Power Supplies

- VDDSRC/VSS: Source voltage domain power supply (Primary Power Rail)
- VDDDST/VSS: Destination voltage domain power supply

Figure 3-3 Level Shifter Cells

3.2.3.2 LSSCL1 Cell

Function

It is a bi-directional voltage level shifter with isolation and is capable of shifting from low-to-high as well as high-to-low. It has to be placed in source domain. When control signal NSLEEP is active, the output is clamped to logic HIGH in output voltage domain.

Power Supplies

- VDDSRC/VSS: Source voltage domain power supply (Primary Power Rail)
- VDDST/VSS: Destination voltage domain power supply

3.2.3.3 LSSCL0 Cell

Function

It is a bi-directional voltage level shifter with isolation and is capable of shifting from low-to-high as well as high-to-low. It has to be placed in source domain. When control signal NSLEEP is active, the output is clamped to logic LOW in output voltage domain.

Power Supplies

- VDDSRC/VSS: Source voltage domain power supply (Primary Power Rail)
- VDDDST/VSS: Destination voltage domain power supply.

3.2.3.4 LSD Cell

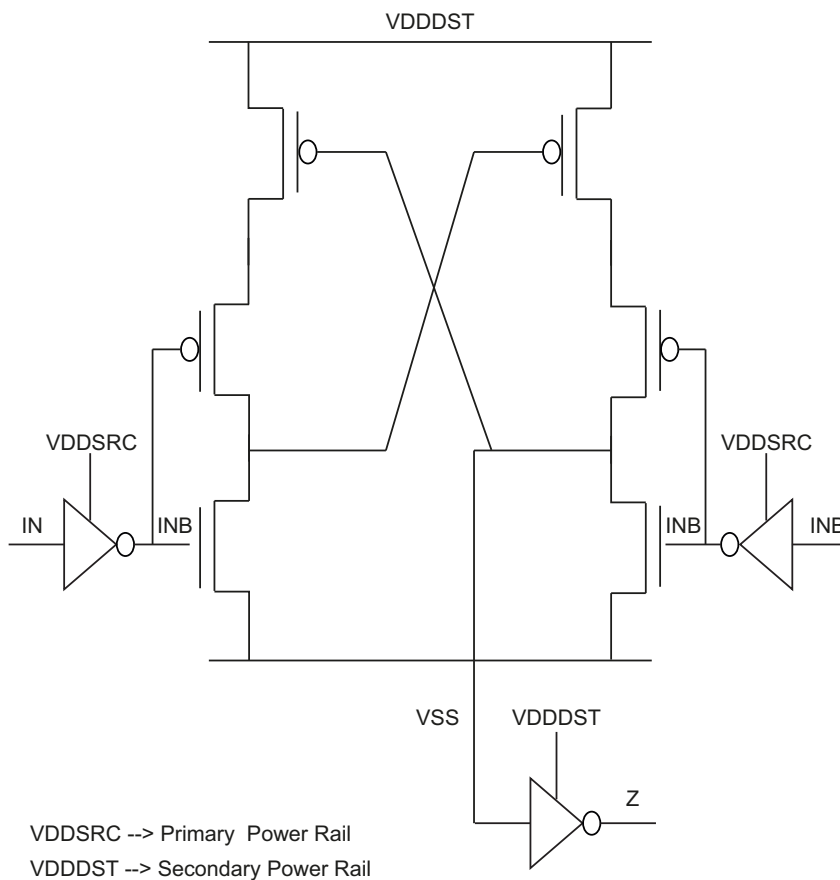
Function

It is a bi-directional voltage level shifter that is capable of shifting from low-to-high as well as high-to-low. It has to be placed in destination domain.

Power Supplies

- VDDSRC/VSS: Source voltage domain power supply
- VDDDST/VSS: Destination voltage domain power supply (Primary Power Rail)

Figure 3-4 LSD Cell



3.2.3.5 LSDCL1 Cell

Function

It is a bi-directional voltage level shifter with isolation and is capable of shifting from low-to-high as well as high-to-low. It has to be placed in destination domain. When control signal NSLEEP is active, the output is clamped to logic HIGH in output voltage domain.

Power Supplies

- VDDSRC/VSS: Source voltage domain power supply
- VDDDST/VSS: Destination voltage domain power supply (Primary Power Rail)

3.2.3.6 LSDCL0 Cell

Function

It is a bi-directional voltage level shifter with isolation and is capable of shifting from low-to-high as well as high-to-low. It has to be placed in destination domain. When control signal NSLEEP is active, the output is clamped to logic LOW in output voltage domain.

Power Supplies

- VDDSRC/VSS: Source voltage domain power supply
- VDDDST/VSS: Destination voltage domain power supply (Primary Power Rail)

3.2.3.7 LSSHL Cell

Function

It is a voltage level shifter that is capable of shifting from high-to-low only. It has to be placed in source domain.

Power Supplies

- VDDSRC/VSS: Source voltage domain power supply (Primary Power Rail)
- VDDDST/VSS: Destination voltage domain power supply

3.2.3.8 LSSHLCL1 Cell

Function

It is a voltage level shifter with isolation that is capable of shifting from high-to-low only. It has to be placed in source domain. When control signal NSLEEP is active, the output is clamped to logic HIGH in output voltage domain.

Power Supplies

- VDDSRC/VSS: Source voltage domain power supply (Primary Power Rail)
- VDDDST/VSS: Destination voltage domain power supply

3.2.3.9 LSSHLCL0 Cell

Function

It is a voltage level shifter with isolation and is capable of shifting from high-to-low only. It has to be placed in source domain. When control signal NSLEEP is active, the output is clamped to logic LOW in output voltage domain.

Power Supplies

- VDDSRC/VSS: Source voltage domain power supply (Primary Power Rail)
- VDDDST/VSS: Destination voltage domain power supply

3.2.3.10 LSDHL Cell

Function

It is a voltage level shifter that is capable of shifting from high-to-low only. It has to be placed in destination domain.

Power Supplies

VDDDST/VSS: Destination voltage domain power supply (Primary Power Rail)

3.2.3.11 LSDHLCL1 Cell

Function

It is a voltage level shifter with isolation and is capable of shifting from high-to-low only. It has to be placed in destination domain. When control signal NSLEEP is active, the output is clamped to logic HIGH in output voltage domain.

Power Supplies

VDDDST/VSS: Destination voltage domain power supply (Primary Power Rail)

3.2.3.12 LSDHLCL0 Cell

Function

It is a voltage level shifter with isolation and is capable of shifting from high-to-low only. It has to be placed in destination domain. When control signal NSLEEP is active, the output is clamped to logic LOW in output voltage domain.

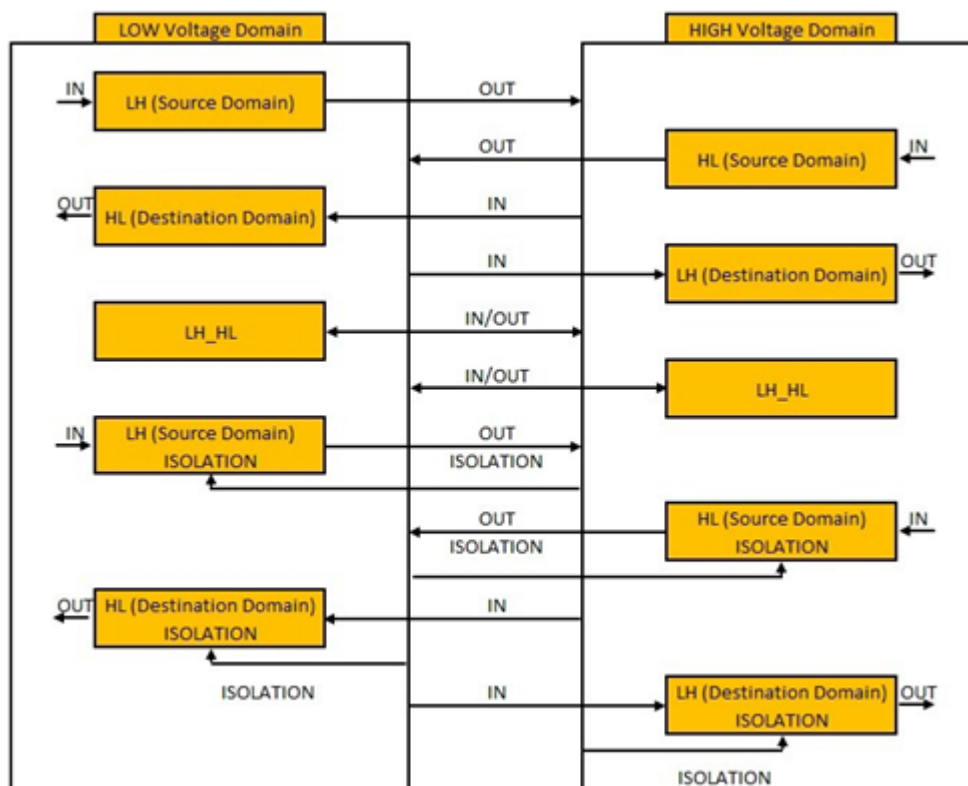
Power Supplies

VDDDST/VSS: Destination voltage domain power supply (Primary Power Rail)

3.2.4 Usage Scenario of Level Shifter Cells

The following figure shows the usage scenario of the level shifter cells.

Figure 3-5 Usage Scenario of Level Shifter Cells



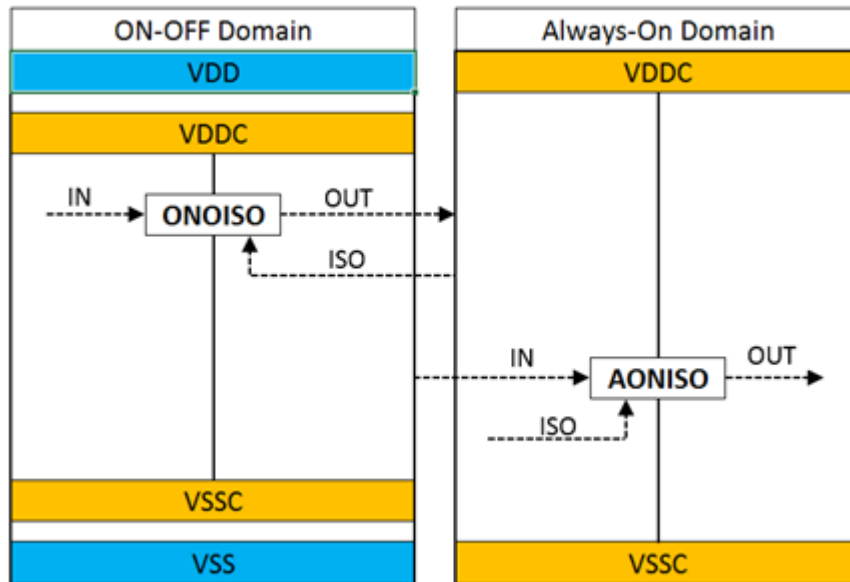
3.3 Power Shut Off (PSO)

Some blocks can be power-gated to reduce power consumption. But the signals crossing from OFF blocks (Switchable Domain) to ON blocks (Always-On Domain) need to be tied to know the logic values like 0/1 to avoid functional failures in Always-On (AON) domains.

3.3.1 Isolation Cell

Isolation cells are placed in ONO and AON domain for isolating the output, and for tying it to Logic 0 or Logic 1 based on the cell used, in case ONO domain is powered down.

Figure 3-6 Isolation Cell



3.3.1.1 AONISOHIH

Function

The cell is placed in AON domain. When control signal ISO is active (HIGH), output gets tied to Logic 1.

Power Supplies

VDDC/VSSC: Always-On power supply/ground

3.3.1.2 AONISOLOH

Function

The cell is placed in AON domain. When control signal ISO is active (HIGH), output gets tied to Logic 0.

Power Supplies

VDDC/VSSC: Always-On power supply/ground

3.3.1.3 ONOISOHI

Function

The cell is placed in On-Off domain. When control signal ISO is active (HIGH), output gets tied to Logic 1.

Power Supplies

- VDD/VSS: Switchable power supply/ground
- VDDC/VSSC: Always-On power supply/ground

3.3.1.4 AONISOLOH

Function

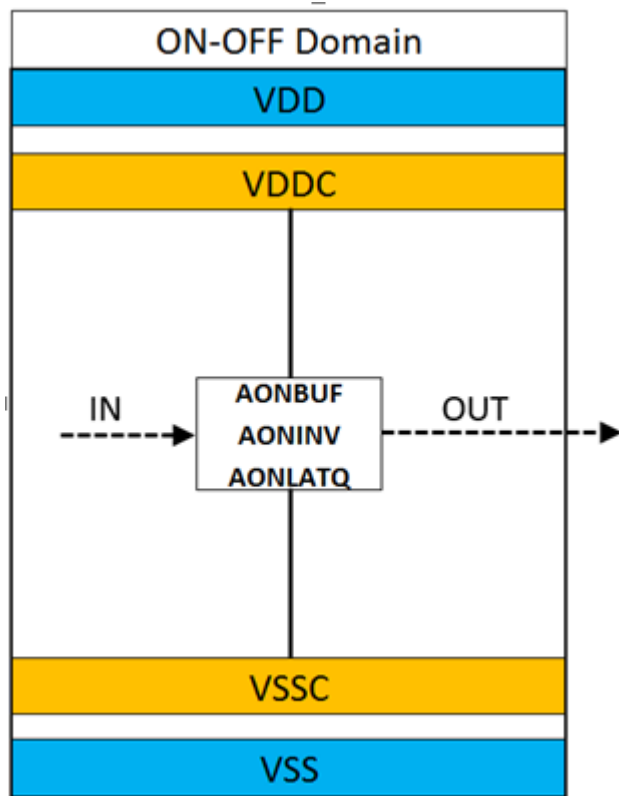
The cell is placed in On-Off domain. When control signal ISO is active (HIGH), output gets tied to Logic 0.

Power Supplies

- VDD/VSS: Switchable power supply/ground
- VDDC/VSSC: Always-On power supply/ground

3.3.2 Always-On Cell

AON cells are placed in ONO domain to drive signals through them, in case ONO domain is powered down.

Figure 3-7 Always-On Cell**3.3.2.1 AONBUF****Function**

The cell is placed in On-Off domain. The cell functions as buffer, driven by AON supply.

Power Supplies

- VDD/VSS: Switchable power supply/ground
- VDDC/VSSC: Always-On power supply/ground

3.3.2.2 AONINV**Function**

The cell is placed in On-Off domain. The cell functions as inverter, driven by AON supply.

Power Supplies

- VDD/VSS: Switchable power supply/ground
- VDDC/VSSC: Always-On power supply/ground

3.3.2.3 AONLATQ

Function

The cell is placed in On-Off domain. The cell functions as active HIGH transparent latch, driven by Always-On supply.

Power Supplies

- VDD/VSS: Switchable power supply/ground
- VDDC/VSSC: Always-On power supply/ground

3.3.3 Power Switch Cells

Header and Footer switch cells are intended for power and ground gating respectively, to avoid the huge rush current because of massive parallel switching during power-up mode.

The following are the primary electrical characteristics of power switches.

- Ion: Current flowing through the switch in ON state, considering a voltage drop of 10 mV between VDDC and VDD.
- Ron: Device resistance of the switch in ON state, considering a voltage drop of 10 mV between VDDC and VDD.
- Rush Current: Current flowing through the switch, considering a maximum voltage drop of VDD.

3.3.3.1 HEADER

Function

To gate the Power supply of an On-Off domain, controlled by signal EN which is active LOW. The cell is to be placed in On-Off domain.

Power Supplies

- VDDC: Always-On power supply
- VDD: Generated/Switchable power supply

3.3.3.2 HEADERBUF

Function

To gate the power supply of an On-Off domain, controlled by signal EN which is active LOW. The cell is to be placed in On-Off domain. This cell has an additional buffered output for control signal EN, which is used to connect the multiple switch cells in a daisy chain manner to reduce the rush currents.

Power Supplies

- VDDC: Always-On power supply
- VDD: Generated/Switchable power supply

3.3.3.3 HEADER - Mother/Daughter Architecture

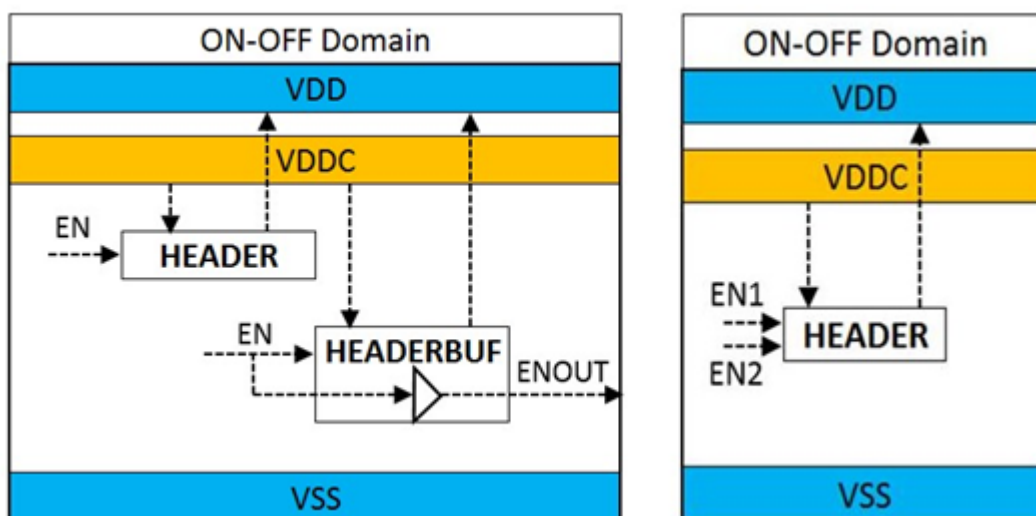
Function

To gate the power supply of an On-Off domain, controlled by signals EN1 and EN2 which are active LOW. The smaller switch is turned on first to get the voltage up to 95%, then the bigger switch is turned on to reduce the IR drop. The cell is to be placed in On-Off domain.

Power Supplies

- VDDC: Always-On power supply
- VDD: Generated/Switchable power supply

Figure 3-8 HEADER - Mother/Daughter Architecture



The following tables represent the electrical characteristics of the different Header style power switches available in the offering for standard channel length (20 nm) SLVT & RVT, simulated at the FFG_0P88V_0P00V_0P00V_0P00V_125C PVT condition

Table 3-3 Electrical Characteristics of Header Style Power Switches - SLVT

Header Cells	I _{on} (μA)	R _{on} (?)	I _{rush} (A)	I _{rushOffVdd} (μA)
SC8T2V_HEADER200X22_CSC20SL	198.9818	50.25585	0.00431338	11.2
SC8T4V_HEADER400X44_CSC20SL	402.485	24.84565	0.00886726	23
SC8T4V_HEADER600X68_CSC20SL	595.2121	16.80074	0.01377718	36.7
SC8T2V_HEADERBUF200X28_CSC20SL	197.8817	50.53524	0.0043119	11.2
SC8T4V_HEADERBUF400X52_CSC20SL	397.2486	25.17315	0.00884748	22.7
SC8T4V_HEADERBUF600X80_CSC20SL	592.5972	16.87487	0.01374366	35.5

Table 3-4 Electrical Characteristics of Header Style Power Switches - RVT

Header Cells	Ion(μA)	Ron(Ω)	Irush(A)	IrushOffVdd(μA)
SC8T2V_HEADER200X22_CSC20R	181.2423355	55.04341183	0.003782924	4.12177344
SC8T4V_HEADER400X44_CSC20R	371.2309764	26.83502421	0.007781814	8.455472819
SC8T4V_HEADER600X68_CSC20R	563.3274513	17.67837914	0.012092541	13.30060246
SC8T2V_HEADERBUF200X28_CSC20R	180.9650656	55.22121205	0.003778452	4.543164605
SC8T4V_HEADERBUF400X52_CSC20R	370.4436694	26.91682282	0.007758494	8.753682002
SC8T4V_HEADERBUF600X80_CSC20R	563.3274513	17.67837914	0.012092541	1.33006E-05

Table 3-5 Electrical Characteristics of Mother-Daughter Header Style Power Switches - SLVT

Header Cells	Ion (Daughter) (μA)	Ion (Mother) (μA)	Ron (Daughter) (Ω)	Ron (Mother) (Ω)	Irush (μA)
SC8T2V_HEADERM200D1X22_CSC20SL	11.4	198.9197	875.0973	50.27154	4535.334
SC8T2V_HEADERM300D1X34_CSC20SL	11.6	307.1434	859.6574	32.55808	7174.326
SC8T4V_HEADERM400D2X44_CSC20SL	22.4	392.5394	445.573	25.47515	9089.362
SC8T_HEADERM100D1X13_CSC20SL	11.5	104.8878	868.3432	95.33998	2496.908

Table 3-6 Electrical Characteristics of Mother-Daughter Header Style Power Switches - RVT

Header Cells	Ion (Daughter) (μA)	Ion (Mother) (μA)	Ron (Daughter) (Ω)	Ron (Mother) (Ω)	Irush (μA)
SC8T2V_HEADERM200D1X22_CSC20R	10.16131181	181.2339224	1096.219578	55.04579622	3978.570954
SC8T2V_HEADERM300D1X34_CSC20R	10.26425309	285.2922454	1091.325834	34.93371522	6293.540341
SC8T4V_HEADERM400D1X44_CSC20R	10.42439755	371.2430355	1077.958071	26.83408076	7977.601417
SC8T_HEADERM100D1X13_CSC20R	10.26181618	95.2792983	1083.293289	104.9828665	2188.508523

3.3.3.4 FOOTER

Function

To gate the Ground supply of an On-Off domain, controlled by signal EN which is active low. The cell is to be placed in On-Off domain.

Power Supplies

- VSSC: Always-On ground supply
- VSS: Generated/Switchable ground supply

3.3.3.5 FOOTERBUF

Function

To gate the Ground supply of an On-Off domain, controlled by signal EN which is active low. The cell is to be placed in On-Off domain. This cell has an additional buffered output for control signal EN, which is used to connect the multiple switch cells in a daisy chain manner to reduce the rush currents.

Power Supplies

- VSSC: Always-On ground supply
- VSS: Generated/Switchable ground supply

3.4 Power Shut Off with State Retention

Some blocks can be power gated to reduce the power, but need to retain the logic during the power shut down mode which can be restored when block comes back to power-up mode. The following figure represents the typical Power-Down and Power-Up sequence of retention cells.

Figure 3-9 Typical Power-Down and Power-Up Sequence of Retention Cells (RET1CSDF*)

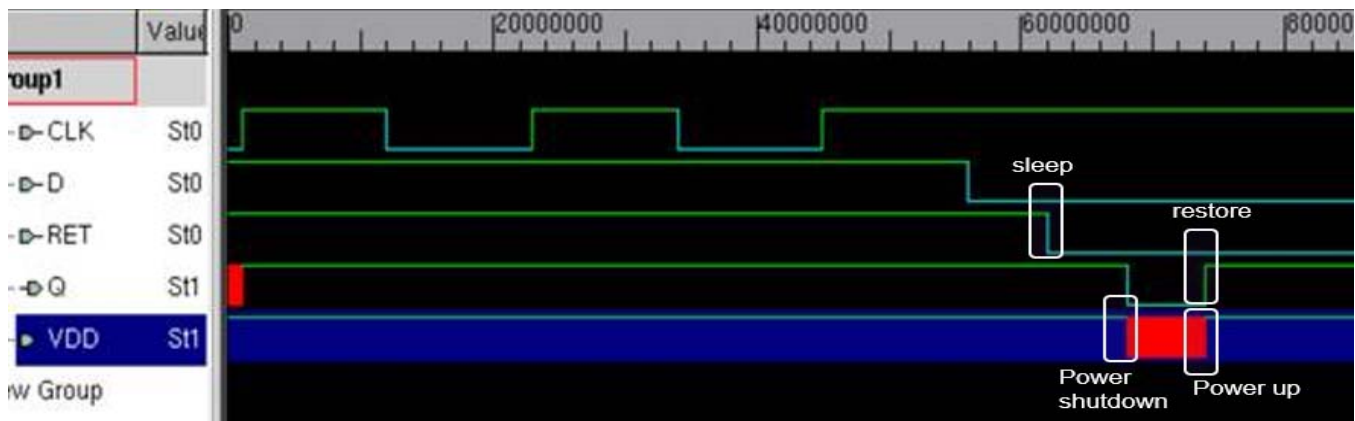
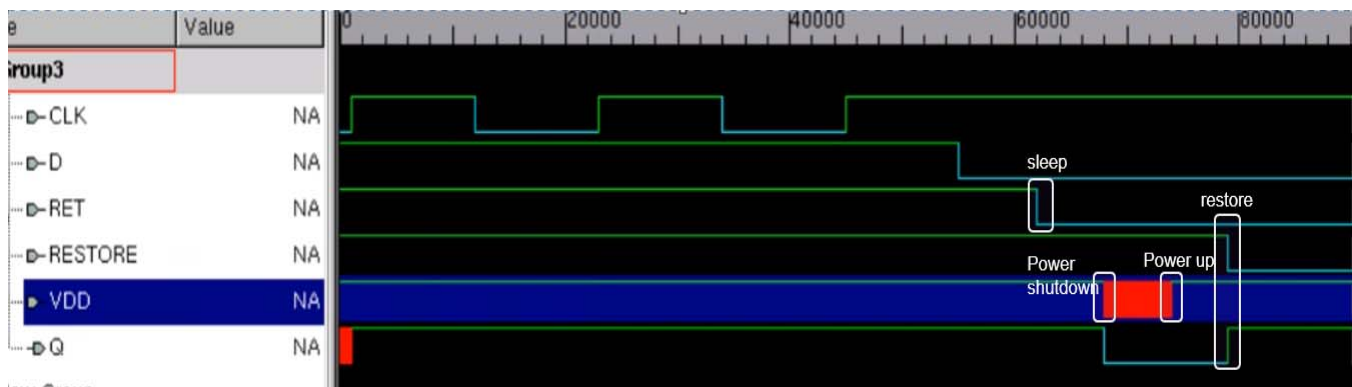


Figure 3-10 Typical Power-Down and Power-Up Sequence of Retention Cells (RET2CSDFF*)

3.4.1 RET1CSDFFQ Cell

Function

Scan flop with Q output with retention logic. Extra shadow latch operates on Always-On power supply (VDDC) to retain the last logic state. When RET signal is active (HIGH) the flop enters Retention mode when VDD is powered off and restores the data as soon as VDD is powered on. Correct save & restore operation requires CLK to be in the active state (HIGH).

Power Supplies

- VDD: Switchable power supply
- VDDC: Always-On power supply
- VSS: Ground

3.4.2 RET1CSDFFRQ Cell

Function

Scan flop with Q output, having asynchronous RESET, with retention logic. Extra shadow latch operates on Always-On power supply (VDDC) to retain the last logic state. When RET signal is active (HIGH), the flop enters Retention mode when VDD is powered off and restores the data as soon as VDD is powered on. RESET signal has priority over RET signal in Functional mode. Correct save & restore operation requires CLK to be in the active state (HIGH).

Power Supplies

- VDD: Switchable power supply
- VDDC: Always-On power supply
- VSS: Ground

3.4.3 RET1CSDFFSQ Cell

Function

Scan flop with Q output, having asynchronous SET, with retention logic. Extra shadow latch operates on Always-On power supply (VDDC) to retain the last logic state. When RET signal is active (HIGH) the flop enters Retention mode when VDD is powered off, and restores the data as soon as VDD is powered on. SET signal has priority over RET signal, in functional mode. Correct save & restore operation requires CLK to be in the active state (HIGH).

Power Supplies

- VDD: Switchable power supply
- VDDC: Always-On power supply
- VSS: Ground

3.4.4 RET2CSDFQ Cell

Function

Scan flop with Q output with retention logic. Extra shadow latch operates on Always-On power supply (VDDC) to retain the last logic state. When RET signal is active (HIGH) the flop enters Retention mode, when VDD is powered off. The saved logic restores when RESTORE signal become active (LOW) and switchable power supply (VDD) is powered on. Correct save & restore operation requires CLK to be in the active state (HIGH).

Power Supplies

- VDD: Switchable power supply
- VDDC: Always-On power supply
- VSS: Ground

3.4.5 RET2CSDFRQ Cell

Function

Scan flop with Q output, having asynchronous RESET, with retention logic. Extra shadow latch operates on Always-On power supply (VDDC) to retain the last logic state. When RET signal is active (HIGH), the flop enters Retention mode when VDD is powered off, and the saved logic restores when RESTORE signal become active (LOW) and switchable power supply (VDD) is powered on. RESET signal has priority over RET signal, in Functional mode. Correct save & restore operation requires CLK to be in the active state (HIGH).

Power Supplies

- VDD: Switchable power supply
- VDDC: Always-On power supply

-
- VSS: Ground

3.4.6 RET2CSDFFSQ Cell

Function

Scan flop with Q output, having asynchronous SET, with retention logic. Extra shadow latch operates on Always-On power supply (VDDC) to retain the last logic state. When RET signal is active (HIGH) the flop enters Retention mode when VDD is powered off, and the saved logic restores when RESTORE signal become active (LOW) and switchable power supply (VDD) is powered on. SET signal has priority over RET signal, in Functional mode. Correct save & restore operation requires CLK to be in the active state (HIGH).

Power Supplies

- VDD: Switchable power supply
- VDDC: Always-On power supply
- VSS: Ground

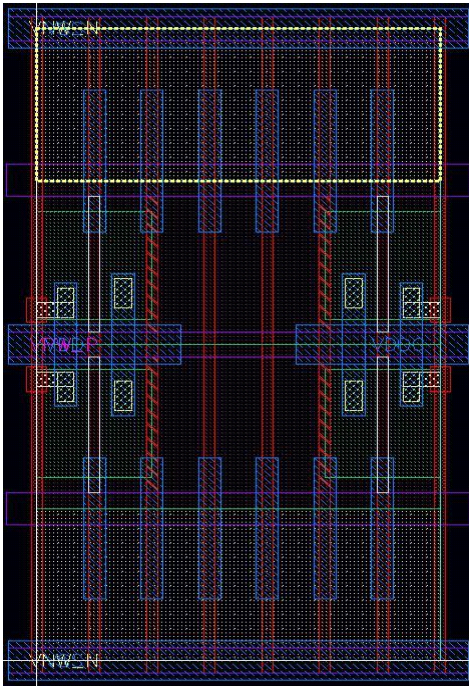
3.5 Physical Cells - Spacer and Tap Cells

If any Always-On cell needs to be placed in CORE row, we can break power rails using spacer cells.

**Note**

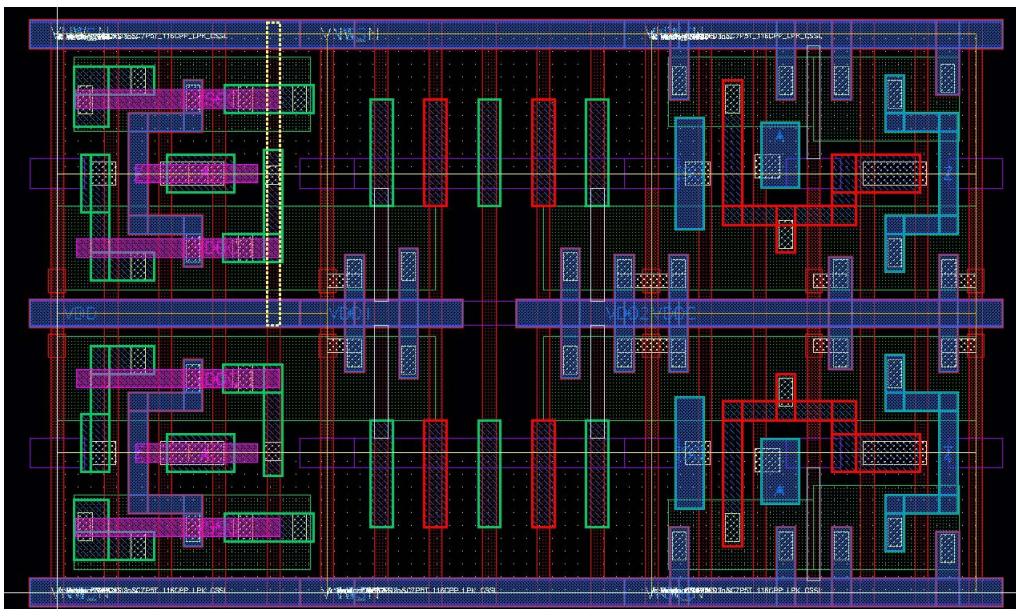
We can also break wells in the same cells, if required, but there is no well breaking in 22FDX

The following figure shows the layout snapshot of spacer cells.

Figure 3-11 Physical Cells - Spacer Cells - 1

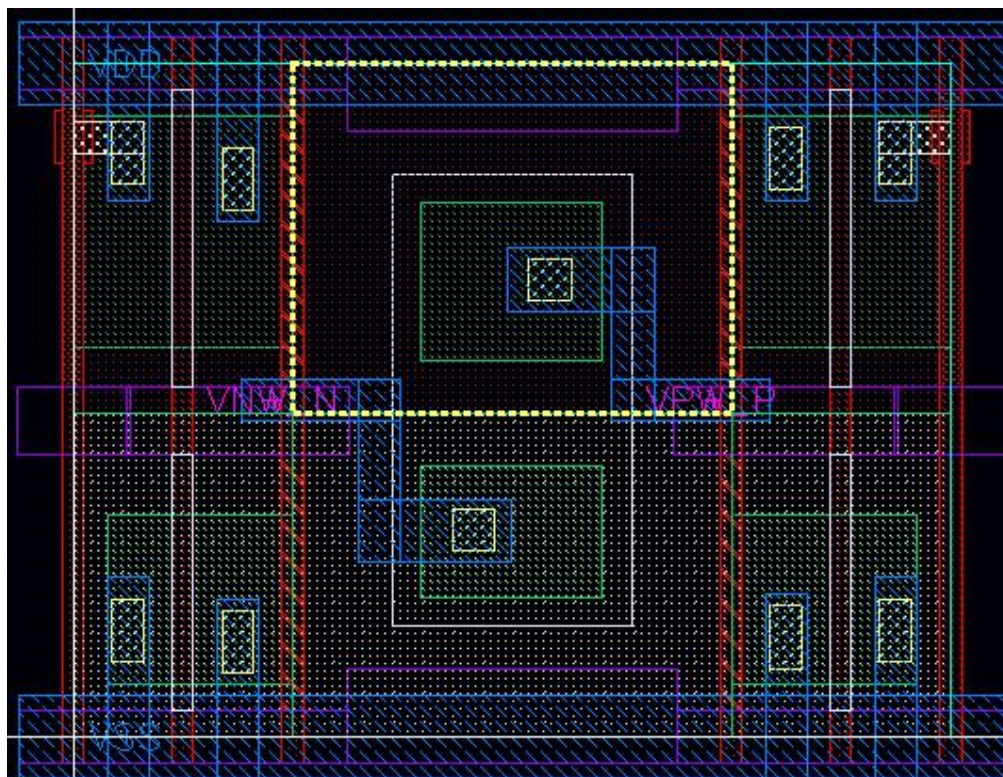
Usage: Break the power rails.

In the following example, VDD and VDDC are separated using spacer cell.

Figure 3-12 Physical Cells - Spacer Cells - 2

As there is no well breaking concept in 22FDX, there is no need to have separate AONTAP cell. The following figure shows the TAP cells.

Figure 3-13 Physical Cells - TAP Cells



4

High Performance Kit (HPK)

4.1 Summary of Supported Cells and Intended SoC Level Usage

The HPK kit contains the differentiated cells to be used along with Base in regular RTL-to-GDS flow to improve the PPA at SoC level. The following table shows the summary of cell types used and their intended use at SoC level.

Table 4-1 Summary of Supported Cells and Intended SoC Level Usage

Cell Name	Cell Type	Usage	Cell Level PPA Gain Compared to Base
C*	Critical Path Beta, Optimized for one Input Arc Keeping same C_{in}	Useful for critical path to reduce WNS	~ 7% – 8% Improvement on cost of ~ 15 area
*SDFF*_HP*	High Performance flops	Useful for achieving High Performance in critical Path	~ 30% Performance Gain @ ~ 30% Area cost
(2B/ 4B) SDFF	Multi Bit Flops 2- bit and 4- bit	To save area and power	~10% – 15% lesser area and switching power
SYNC	Clock Synchronizer Flops 2-stage, 3-stage, and 4-stage	To be used between two clock domain crossings	Optimized for better MTBF
*SDFF*_LP_	Low Power Flops	Useful to achieve low cell internal power in power critical paths	~10% improvement in cell level internal power
R*	Skewed Beta, Optimized for rise arc by reducing C_{in} in rise arc	Used for power recovery and to reduce TNS/WNS	5%–10% improvement in rise arc C_{in}
F*	Skewed Beta, Optimized for fall arc by reducing C_{in} in fall arc	Used for power recovery and to reduce TNS/WNS	5%–10% improvement in fall arc C_{in}

4.2 Critical Path Beta Cells

Cells with improved delay, at the cost of area degradation, are to be used in the critical paths of the design to push frequency. Same input cap for one of the in-out arcs with improved rise and fall delays. Other in-out arc will have more input cap and cell area will be more compared to base versions.

4.3 High Performance Flop

HP flops are designed to meet the high performance goals of the design by improving clk2q and improved setup time compared to conventional master-slave flip flop. Due to separate path for Data and Scan input setup time improvement can be achieved by up to 30%–35% and CLK2Q delay by about 25%–30% with an area increase by ~30%. Constraining the max transition to ~10% of clock period at design level will yield better PPA.

4.4 Multi-bit Flops

Multi-bit flip-flop is an effective power saving methodology by merging two single bit scan flops in the design. The Multi-bit flip flops can save clock dynamic power and the area of flop effectively.

4.4.1 2BSDFFQ Cell

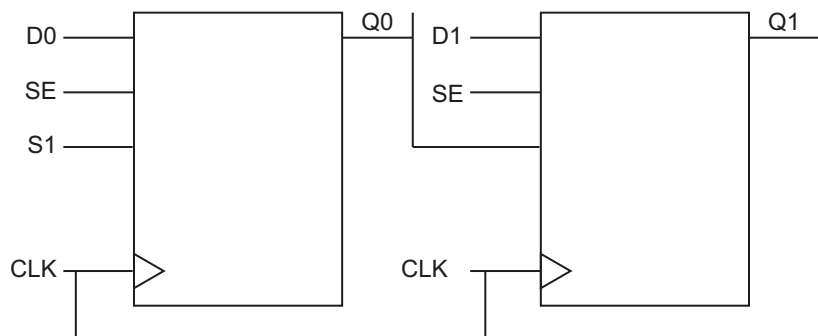
Function

Two single bit scan flops with Q output, combined by sharing the clock buffers. The scan chain is internally stitched with scan input pin being SI and scan enable SE being shared across all the flops.

Pins

- Inputs: D0, D1, SI, SE, CLK
- Outputs: Q0, Q1

Figure 4-1 2BSDFFQ Cell



4.4.2 4BSDFFQ Cell

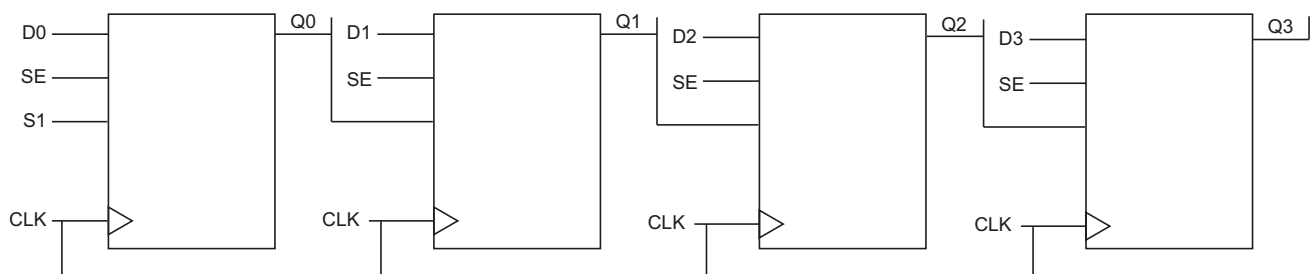
Function

Four single bit scan flops with Q output, combined by sharing the clock buffers. The scan chain is internally stitched with scan input pin being SI and scan enable SE being shared across all the flops.

Pins

- Inputs: D0, D1, D2, D3, SI, SE, CLK
- Outputs: Q0, Q1, Q2, Q3

Figure 4-2 4BSDFFQ Cell



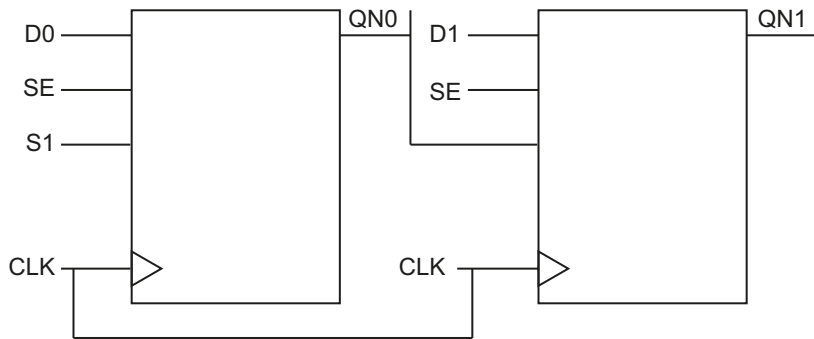
4.4.3 2BSDFFQN Cell

Function

Two single bit scan flops with QN output, combined by sharing the clock buffers. The scan chain is internally stitched with scan input pin being SI and scan enable SE being shared across all the flops.

Pins

- Inputs: D0, D1, SI, SE, CLK
- Outputs: QN0, QN1

Figure 4-3 2BSDFFQN Cell

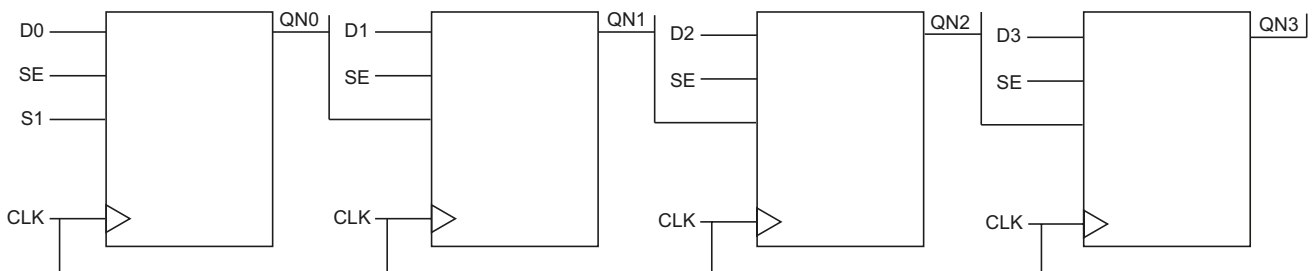
4.4.4 4BSDFFQN Cell

Function

Four single bit scan flops with QN output, combined by sharing the clock buffers. The scan chain is internally stitched with scan input pin being SI and scan enable SE being shared across all the flops.

Pins

- Inputs: D0, D1, D2, D3, SI, SE, CLK
- Outputs: QN0, QN1, QN2, QN3

Figure 4-4 4BSDFFQN Cell

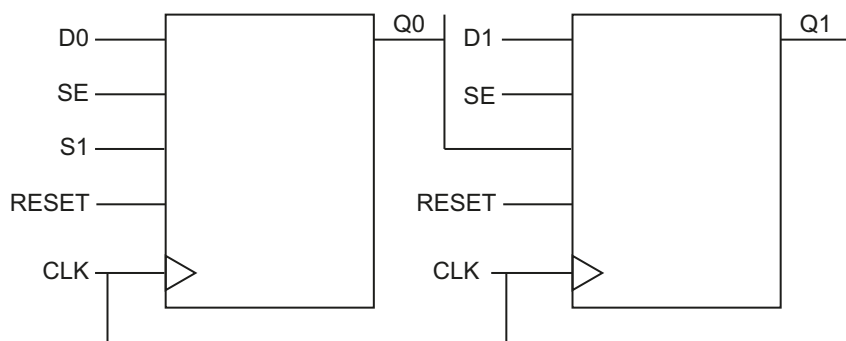
4.4.5 2BSDFFRQ Cell

Function

Two single bit scan flops with Q output and asynchronous RESET, combined by sharing the clock buffers. The scan chain is internally stitched with scan input pin being SI and scan enable SE being shared across all the flops. RESET pin is also shared across all the flops.

Pins

- Inputs: D0, D1, SI, SE, RESET, CLK
- Outputs: Q0, Q1

Figure 4-5 2BSDFFRQ Cell

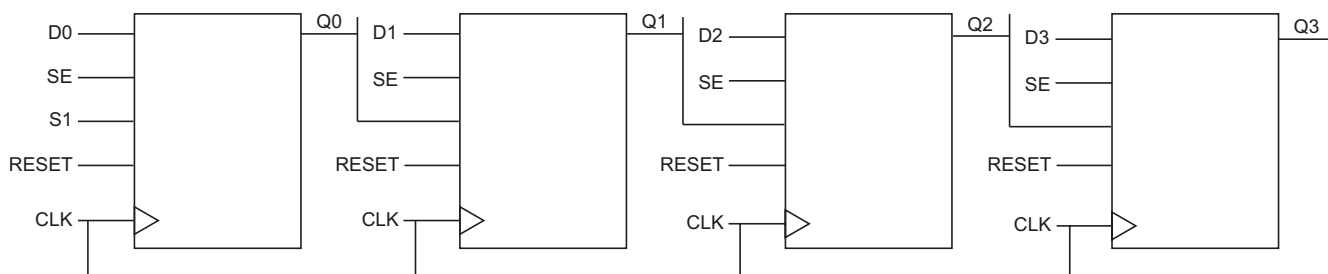
4.4.6 4BSDFFRQ Cell

Function

Four single bit scan flops with Q output and asynchronous RESET, combined by sharing the clock buffers. The scan chain is internally stitched with scan input pin being SI and scan enable SE being shared across all the flops. RESET pin is also shared across all the flops.

Pins

- Inputs: D0, D1, D2, D3, SI, SE, RESET, CLK
- Outputs: Q0, Q1, Q2, Q3

Figure 4-6 4BSDFFRQ Cell

4.4.7 2BSDFFRQN Cell

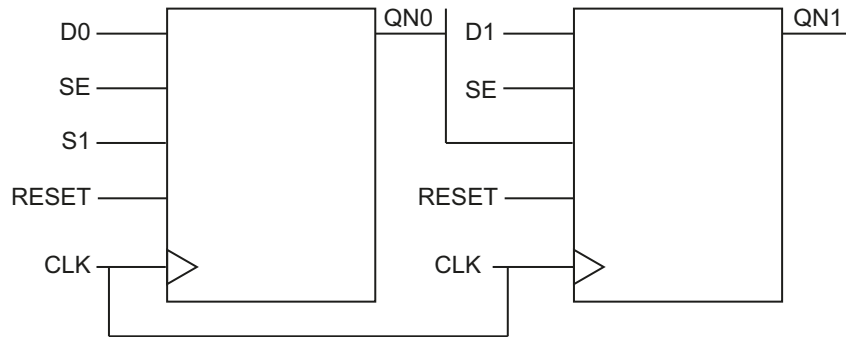
Function

Two single bit scan flops with QN output and asynchronous RESET, combined by sharing the clock buffers. The scan chain is internally stitched with scan input pin being SI and scan enable SE being shared across all the flops. RESET pin is also shared across all the flops.

Pins

- Input: D0, D1, SI, SE, RESET, CLK

- Output: QN0, QN1

Figure 4-7 2BSDFFRQN Cell

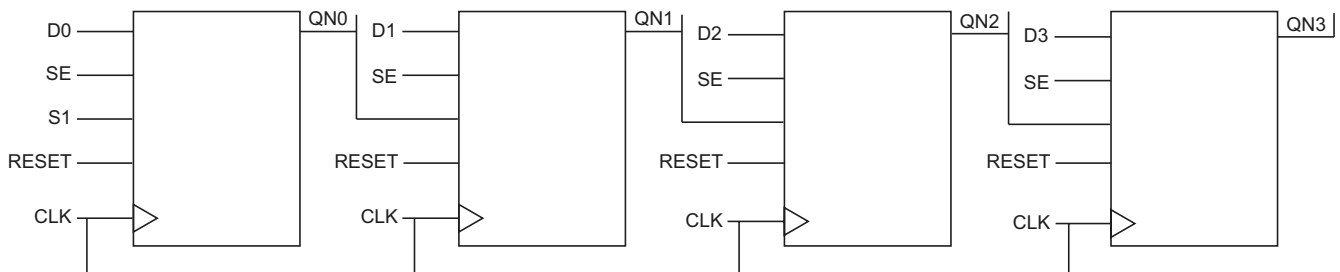
4.4.8 4BSDFFRQN Cell

Function

Four single bit scan flops with QN output and asynchronous RESET, combined by sharing the clock buffers. The scan chain is internally stitched with scan input pin being SI and scan enable SE being shared across all the flops. RESET pin is also shared across all the flops.

Pins

- Inputs: D0, D1, D2, D3, SI, SE, RESET, CLK
- Outputs: QN0, QN1, QN2, QN3

Figure 4-8 4BSDFFRQN Cell

4.4.9 2BSDFFSQ Cell

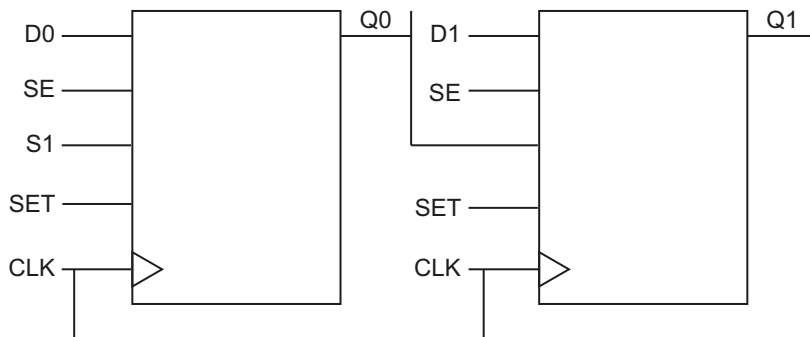
Function

Two single bit scan flops with Q output and asynchronous SET, combined by sharing the clock buffers. The scan chain is internally stitched with scan input pin being SI and scan enable SE being shared across all the flops. RESET pin is also shared across all the flops.

Pins

- Inputs: D0, D1, SI, SE, SET, CLK
- Outputs: Q0, Q1

Figure 4-9 2BSDFFSQ Cell



4.4.10 4BSDFFSQ Cell

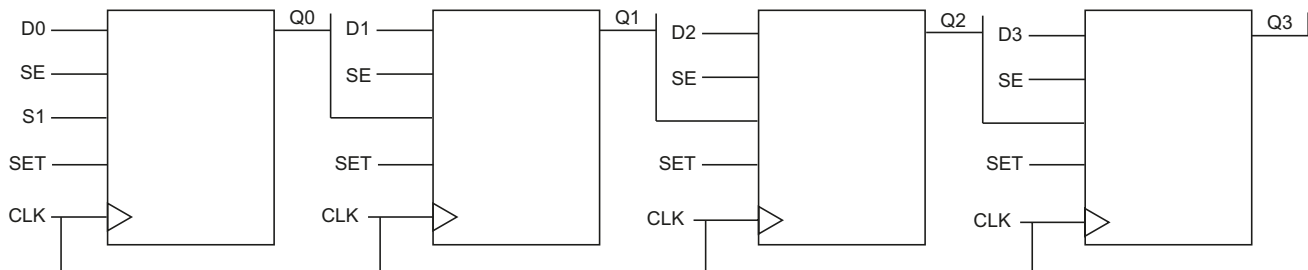
Function

Four single bit scan flops with Q output and asynchronous SET, combined by sharing the clock buffers. The scan chain is internally stitched with scan input pin being SI and scan enable SE being shared across all the flops. RESET pin is also shared across all the flops.

Pins

- Inputs: D0, D1, D2, D3, SI, SE, SET, CLK
- Outputs: Q0, Q1, Q2, Q3

Figure 4-10 4BSDFFSQ Cell



4.4.11 2BSDFFSQN Cell

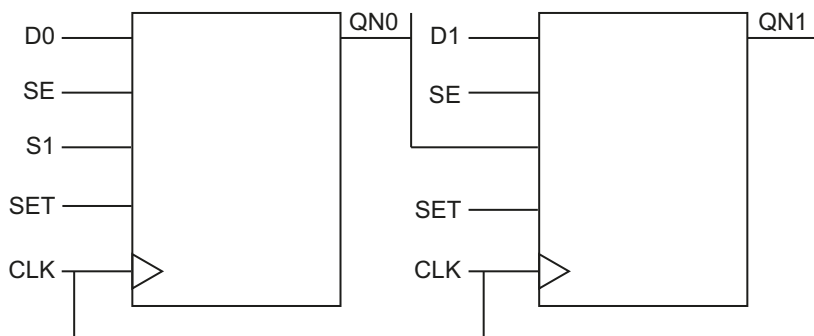
Function

Two single bit scan flops with QN output and asynchronous SET, combined by sharing the clock buffers. The scan chain is internally stitched with scan input pin being SI and scan enable SE being shared across all the flops. RESET pin is also shared across all the flops.

Pins

- Inputs: D0, D1, SI, SE, SET, CLK
- Outputs: QN0, QN1

Figure 4-11 2BSDFFSQN Cell



4.4.12 4BSDFFSQN Cell

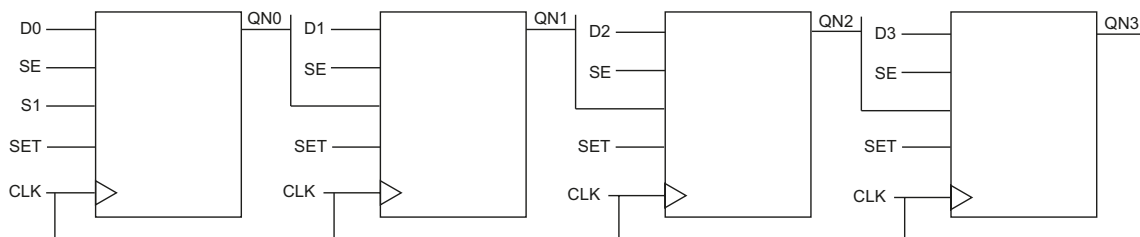
Function

Four single bit scan flops with QN output and asynchronous SET, combined by sharing the clock buffers. The scan chain is internally stitched with scan input pin being SI and scan enable SE being shared across all the flops. RESET pin is also shared across all the flops.

Pins

- Inputs: D0, D1, D2, D3, SI, SE, SET, CLK
- Outputs: QN0, QN1, QN2, QN3

Figure 4-12 4BSDFFSQN Cell



4.5 Synchronizers

Synchronizers are used on asynchronous interfaces to resolve metastability issues that arise due to the high probability of violating setup/hold times in the capture clock domain. Capture flop may go metastable. The probability of this and the time to resolve the metastability depend on many factors inherent to the capture flop, on environmental conditions (V, T), and on the frequency of clock and data transitions. Therefore, use

of synchronizer cells from the Standard Cells library is recommended. Synchronizers can be built from discrete flops, but special care must be taken to ensure that data path delay and clock skew are minimized.

4.5.1 Observations

- Synchronizers cells made of higher threshold voltage devices provide lesser protection.
- More the number of stages in Synchronizer cells, better would be the Metastability resolution.
- If low leakage is priority, use of higher threshold voltage Synchronizer cell is preferable.
- Metastability resolution is dependent on temperature, voltage, and body-bias. Guidance that follows is typically for worst case conditions. Increase in any of these factors helps resolve metastability events quicker and allow for either high frequencies, or allow for simpler synchronization structures

4.5.2 Guidelines for Synchronizer Selection

Based on design frequency, choose the correct Synchronizer VT, Number of stages, and Lgate from [Table 4-2](#) below. To have sufficient margin, it is recommended to choose a SYNC flop with 10% more than the design frequency. If leakage is not a concern, it is recommended to use fastest VT, Lgate, and more number of stages to have better MTBF.

Table 4-2 Simulation Specifications and Criteria

VT	PVT	BIAS	Slew	Assumption		Passing Criteria
SLVT	SSG_0P72V__0P00V_0P00V_0P00V_125C	0.00 V/0.00 V	Liberty Max Slew	Data Freq = 30% of CLK Frequency	No. of Instances =1000	MTBF > 10 Yrs
LVT	SSG_0P72V_0P00V_0P00V_0P00V_M40C					
RVT	SSG_0P72V_0P00V_0P00V_0P00V_M40C					
HVT	SSG_0P72V_0P00V_0P00V_0P00V_M40C					

Table 4-3 Maximum Supported Frequency

VT Type	Flop Type		Non Set Reset (SDFFQ)			Reset (SDFFRQ)			Set (SDFFSQ)		
	No. of Stages	Lgate (nm)	Drive X1	Drive X2	Drive X4	Drive X1	Drive X2	Drive X4	Drive X1	Drive X2	DriveX 4
Maximum Frequency Supported (MHz)											
SLVT	2	20	1960	1888	1880	1562	1542	1540	1787	1785	1713
		24	1568	1510	1508	1306	1298	1297	1401	1398	1347
		28	1305	1247	1242	1087	1080	1085	1147	1173	1174
	3	20	2920	2994	2993	2509	2511	2504	2687	2685	2742
		24	2353	2418	2417	2063	2079	2074	2132	2130	2185
		28	1985	2054	2053	1743	1750	1743	1810	1813	1867
	4	20	3000	3000	3000	3000	3000	3000	3000	3000	3000
		24	2923	2870	2838	2510	2505	2497	2692	2690	2640
		28	2524	2459	2442	2156	2142	2139	2323	2326	2268
LVT	2	20	246	225	222	219	214	214	239	239	220
		24	170	157	152	150	146	146	157	156	144
		28	113	102	98	103	99	99	107	113	96
	3	20	434	474	470	409	400	402	429	429	463
		24	307	336	335	292	285	287	294	294	321
		28	210	228	225	206	200	200	209	210	230
	4	20	690	643	629	587	571	575	679	680	640
		24	511	472	467	436	424	425	504	506	471
		28	369	338	331	318	309	309	381	383	356

Table 4-3 Maximum Supported Frequency (Continued)

Flop Type			Non Set Reset (SDFFQ)			Reset (SDFFRQ)			Set (SDFFSQ)		
VT Type	No. of Stages	Lgate (nm)	Drive X1	Drive X2	Drive X4	Drive X1	Drive X2	Drive X4	Drive X1	Drive X2	DriveX 4
Maximum Frequency Supported (MHz)											
RVT	2	20	927	883	871	717	699	701	845	847	805
		24	768	725	714	576	559	563	706	706	667
		28	604	559	545	464	448	450	585	584	550
	3	20	1443	1499	1497	1206	1184	1181	1338	1340	1397
		24	1200	1253	1252	988	965	964	1128	1128	1181
		28	968	1026	1020	800	778	778	946	951	993
	4	20	1893	1835	1794	1576	1549	1550	1804	1806	1740
		24	1598	1533	1499	1298	1271	1274	1543	1545	1476
		28	1324	1272	1228	1056	1029	1026	1302	1302	1248
HVT	2	20	55	49	48	69	64	65	81	81	71
		24	35	30	30	45	41	41	56	56	49
		28	23	20	20	30	27	27	40	40	35
	3	20	102	116	116	133	127	128	150	150	169
		24	67	76	76	93	87	87	109	109	123
		28	47	52	52	66	61	61	81	80	91
	4	20	191	170	165	194	187	188	270	271	246
		24	132	116	113	139	132	134	206	207	188
		28	94	83	82	103	97	97	158	159	142

4.5.3 Custom Mixed VT Synchronizers

MixedVT Synchronizer cells have been designed to bridge the gap between the only LVT and only SLVT synchronizer cells. Two-stage cells are available in all 3 channel lengths with RESET, SET and without SET-RESET variants, with base VT being LVT.

4.5.3.1 Guide Lines For Synchronizer Selection

Table 4-4 Simulation Specifications and Criteria for MixedVT Cells

VT	PVT	Bias	Slew	Assumptions		Passing Criteria
MixedVT (LVT+SLVT)	SSG_0P59V_0P00V_0P00V_0P00V_M40C	0.00V/ 0.00V	Liberty Max Slew	Data Freq = 30% of CLK Freq	No. of Instance s = 1000	MTBF > 10 yrs
	SSG_0P72V_0P00V_0P00V_0P00V_125C	0.00V/ 0.00V				

Table 4-5 Maximum Supported Frequency for MixedVT Cells

Flop Type			SYNC2 SDFFQ	SYNC2 SDFFRQ	SYNC2 SDFFSQ
PVT	No. Of Stages	Lgate	Drive X1		
Max Freq Supported (MHz)					
SSG_0P59V_0P00V_0P00V_0P00V_M40C	2	20 nm	466	360	414
		24 nm	343	268	305
		28 nm	251	198	221
SSG_0P72V_0P00V_0P00V_0P00V_125C		20 nm	1530	1247	1384
		24 nm	1377	1070	1224
		28 nm	1153	906	1025

5

Engineering Change Order Kit (ECO)

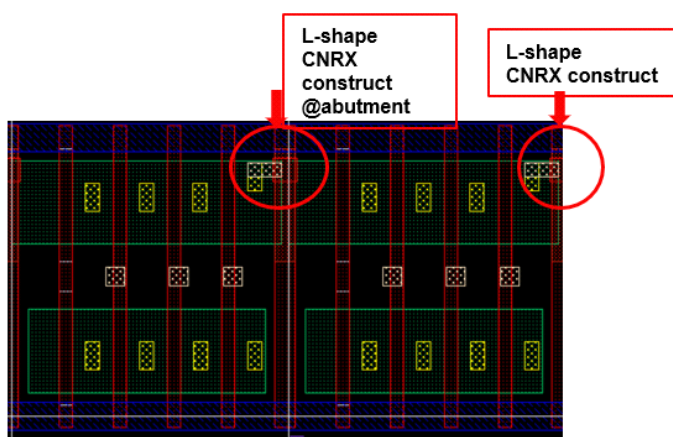
ECO Kit is required to enable fix silicon bugs using metal ECOs or last minute design changes before tape out using only few Metal mask changes. The key benefits are quick turn-around time and significant mask cost saving for fixing the design bugs.

5.1 Architecture

The ECO base cell is of 5CPP wide, and all the other ECO Logic cells, ECO FILL cells, and ECO DECAP cells are implemented using the ECO base cell and changing only Metal layers and keeping same FEOL & MEOL layers. FEOL layers (PC, RX, CT, NW, PPLUS, etc.) and MEOL (CA, CB) used in the ECO base cell are FIXED in position and dimension.

LAYERS in ECOFILL (FIXED positions & dimension)	Additional LAYERS to create ECOLOGIC (No FIXED position &/or dimension)
NW, PP, NP, PC,AUXPC3, CT, *VTN,*VTP,ULPP,ULPN,PCSTP,PCSTN,FLTGATE,CA,CB	M1, M2, V1, (AY,C1 in complex cells)

Figure 5-1 Two ECO Base cells abutted

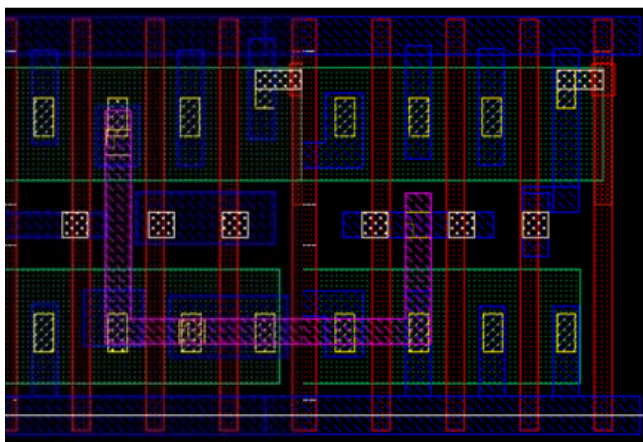


5.2 Design Level Usage

Below is an example on how an AN2X1 cell is created (and part of ECO kit) using two ECO FILL cells or ECO DECAP cells. During regular flow, place the ECO FILL or ECO DECAP cells across the design area. ECO DECAP cells are preferable to ECO FILL cells, as they assist in IR drop reduction with little increase in leakage. To realize the required function during metal ECOs stage, swap the ECO FILL or ECO DECAP cell with ECO logic cell.

ECO cells should be placed only with R0 or Mx orientations.

Figure 5-2 AN2X1 cell implemented using only Metal change from ECO FILL or ECO DECAP cell



5.3 Supported Cells

Nine widths of the ECO filler cells are supported as below. Example, X15 contains three of the 5-CPP ECO FILL cells. This can be used to realize a logic function for which ECO logic cell is of 15 CPP width.

Table 5-1 Supported Cells

Parameters	Values
Min ECO filler width	5 CPP
ECO fillers	X5, X10, X15, X20, X25, X30, X35, X50, X90

All the most used logic functions as per below table are supported in ECO kit. All cells have driven X1, X2 and X4.

Table 5-2 Cell Families Supported in ECO

Cell Families	
1	ECODFFQN
2	ECODFFQ
3	ECODFFRQ
4	ECODFFSQ
5	ECOSDFFQN
6	ECOSDFFQ
7	ECOSDFFRQ
8	ECOSDFFSQ
9	ECOMUx2
10	ECOMUx12
11	ECOAN2
12	ECOA0I21
13	ECOA0I22
14	ECOBUFF
15	ECOxNR2
16	ECOxOR2
17	ECOINV
18	ECOND2
19	ECOND3
20	ECONR2
21	ECONR3
22	ECOOR2
23	ECOAONBUFF
24	ECOAONINV
25	ECOTIEHIGH
26	ECOTIELO

6

Library Modeling

6.1 Timing Delay Model

The timing delay models provided within the library are modeled using look-up table, based on input slew and output capacitive load.

Timing delays are measured with relaxed timing constraints, longer than necessary. The use of tighter timing constraints might unnecessarily increase the delay.

6.1.1 Maximum Transition Time

The following table shows the maximum transition allowed for the 0,0 bias ($V_{Pwell} = V_{Nwell} = 0$ V) PVT sets. The maximum transition allowed for other PVT corners, referring to same VDD value, but different biasing potentials will be same as 0,0 bias PVT corners.

Table 6-1 Maximum Transition Supported for SLVT/LVT Non-Shift Products

PVT	Clock	Data
FFG_0P72V_0P00V_0P00V_0P00V_125C	458.656	948.768
FFG_0P72V_0P00V_0P00V_0P00V_M40C	458.656	948.768
FFG_0P88V_0P00V_0P00V_0P00V_125C	176.411	344.468
FFG_0P88V_0P00V_0P00V_0P00V_M40C	176.411	344.468
FFG_0P945V_0P00V_0P00V_0P00V_125C	141.136	253.392
FFG_0P945V_0P00V_0P00V_0P00V_M40C	141.136	253.392
SSG_0P59V_0P00V_0P00V_0P00V_125C	689.36	1426
SSG_0P59V_0P00V_0P00V_0P00V_M40C	689.36	1426
SSG_0P72V_0P00V_0P00V_0P00V_125C	265.146	517.736
SSG_0P72V_0P00V_0P00V_0P00V_M40C	265.146	517.736
SSG_0P76V_0P00V_0P00V_0P00V_125C	265.146	517.736

Table 6-1 Maximum Transition Supported for SLVT/LVT Non-Shift Products

PVT	Clock	Data
SSG_0P76V_0P00V_0P00V_0P00V_M40C	265.146	517.736
SSG_0P81V_0P00V_0P00V_0P00V_125C	212.12	380.848
SSG_0P81V_0P00V_0P00V_0P00V_M40C	212.12	380.848
TT_0P59V_0P00V_0P00V_0P00V_125C	689.36	1426
TT_0P59V_0P00V_0P00V_0P00V_M40C	689.36	1426
TT_0P65V_0P00V_0P00V_0P00V_25C	484.784	1002.82
TT_0P65V_0P00V_0P00V_0P00V_85C	484.784	1002.82
TT_0P72V_0P00V_0P00V_0P00V_125C	265.146	517.736
TT_0P72V_0P00V_0P00V_0P00V_M40C	265.146	517.736
TT_0P80V_0P00V_0P00V_0P00V_25C	186.46	364.09
TT_0P80V_0P00V_0P00V_0P00V_85C	186.46	364.09
TT_0P81V_0P00V_0P00V_0P00V_125C	212.12	380.848
TT_0P81V_0P00V_0P00V_0P00V_M40C	212.12	380.848
TT_0P90V_0P00V_0P00V_0P00V_25C	149.176	267.824
TT_0P90V_0P00V_0P00V_0P00V_85C	149.176	267.824

Table 6-2 Maximum Transition Supported for RVT/HVT Non-Shift Products

PVT Corner	Clock Max Transition	Data Max Transition
FFG_0P88V_0P00V_0P00V_0P00V_125C	535.784	1009.86
FFG_0P88V_0P00V_0P00V_0P00V_M40C	535.784	1009.86
FFG_0P945V_0P00V_0P00V_0P00V_125C	454.36	846.944
FFG_0P945V_0P00V_0P00V_0P00V_M40C	454.36	846.944
SSG_0P72V_0P00V_0P00V_0P00V_M40C	819.744	1545.09
SSG_0P72V_0P00V_0P00V_0P00V_125C	819.744	1545.09
SSG_0P76V_0P00V_0P00V_0P00V_M40C	819.744	1545.09
SSG_0P76V_0P00V_0P00V_0P00V_125C	819.744	1545.09
SSG_0P81V_0P00V_0P00V_0P00V_M40C	695.176	1295.83
SSG_0P81V_0P00V_0P00V_0P00V_125C	695.176	1295.83

Table 6-2 Maximum Transition Supported for RVT/HVT Non-Shift Products

PVT Corner	Clock Max Transition	Data Max Transition
TT_0P80V_0P00V_0P00V_0P00V_25C	669.728	1262.40
TT_0P80V_0P00V_0P00V_0P00V_85C	669.728	1262.40
TT_0P90V_0P00V_0P00V_0P00V_25C	567.952	1058.68
TT_0P90V_0P00V_0P00V_0P00V_85C	567.952	1058.68

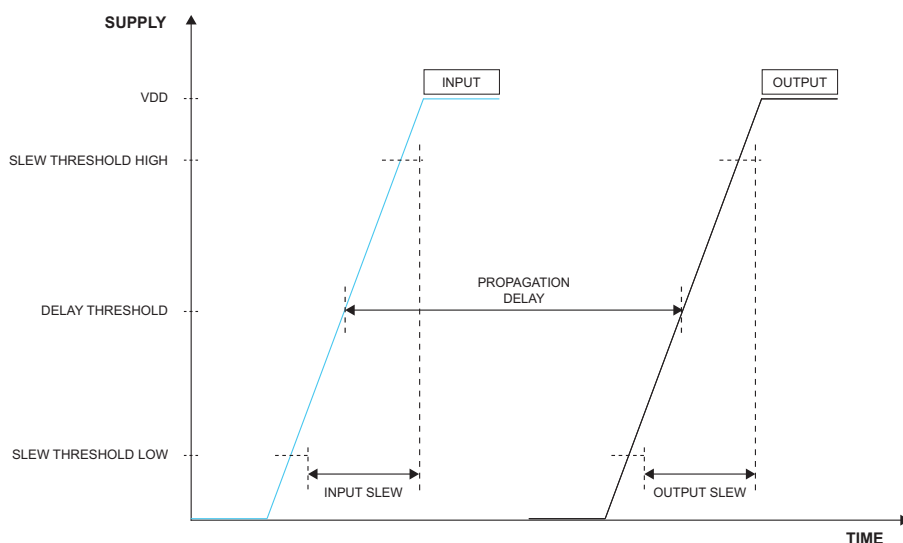
6.1.2 Propagation Delay

The propagation delays are defined as the time interval between the input signal crossing the falling or rising delay thresholds and the output crossing the falling or rising delay thresholds.

6.1.3 Slew / Transition Time

Transition times are defined as the time interval between the signal crossing the upper and lower slew thresholds, based on rising or falling transition.

The following figure shows the calculations of propagation delay and slew.

Figure 6-1 Timing Delay

Note The library might contain negative propagation delays. Most timing verification tools can handle negative propagation delays properly.

6.2 Timing Constraints

Timing constraints define minimum time intervals during which specific signals must be held steady to ensure the correct functioning of sequential cells.

6.2.1 Setup Time

Setup time for a sequential cell represents the minimum time data signal must be stable before the active edge/ level of the clock or other control signal arrives, to ensure correct functionality of cell. Setup constraint values are measured as time interval between the data signal crossing the falling or rising delay threshold point and the clock/ control signal crossing the falling or rising delay threshold point. Setup time is measured with relaxed hold time.

6.2.2 Hold Time

Hold time for a sequential cell represents the minimum time that a data signal must be stable for, after the active edge/ level of the clock or other control signal arrives, to ensure correct functionality of cell. Hold constraint values are measured as time interval between the clock/ control signal crossing the falling or rising delay threshold point, and the data signal crossing the falling or rising delay threshold point. Hold time is measured with relaxed setup time.

**Note**

Synopsys does not add any hold time margins in the timing models. Designers must take care about such margins required depending on their design uncertainties.

6.2.3 Recovery Time

Recovery time for a sequential cell represents the minimum time an asynchronous signal must remain disabled for, before the arrival of active edge of clock signal, to ensure correct functionality of cell. Recovery constraint values are measured as time interval between the asynchronous signal crossing the falling or rising delay threshold point, and the clock signal crossing the falling or rising delay threshold point. Recovery time is measured with relaxed removal time.

6.2.4 Removal Time

Removal time for a sequential cell represents the minimum time an asynchronous signal must remain enabled after the arrival of active edge of clock signal, to ensure correct functionality of cell. Removal constraint values are measured as time interval between the clock signal crossing the falling or rising delay threshold point, and the asynchronous signal crossing the falling or rising delay threshold point. Removal time is measured with relaxed recovery Time.

6.2.5 Minimum Pulse Width

Minimum pulse width for a sequential cell represents the minimum time a signal must remain high or low to ensure correct functionality of cell. Minimum pulse width high constraint value is measured as the time interval between the rising edge of the signal crossing rising delay threshold point and the falling edge of the signal crossing falling delay threshold point. Minimum pulse width low constraint value is measured as

the time interval between the falling edge of the signal crossing falling delay threshold point, and the rising edge of the signal crossing rising delay threshold point.

6.2.6 Power Model

Two forms of power data are modeled primarily in the library. Static power is represented in the form of leakage power, while dynamic power is represented in the form of Internal energy.

6.2.6.1 Leakage Power

Leakage power represents the power dissipation which happens when cell is in a stable state, predominantly contributed by the OFF currents of the devices.

6.2.6.2 Internal Energy

Internal energy represents the energy being consumed within the cell whenever one or more inputs of the cell switch their states at particular process, voltage and temperature. The switching of inputs may or may not result in the switching of concerned outputs, which is primarily a function of the cell topology. The internal energy data in the library is measured for the combinations of input slew and output loads. power dissipation depends on the power-supply voltage, frequency of the operation, internal capacitance, and output load. This internal energy data can be used to estimate the total power dissipation of a cell within a design, along with information regarding the input pin capacitance and design-specific information.

7

Library Usage Guidelines

7.1 Summary of Physical Cells Supported and Usage

7.1.1 Why Different ENDCAP Cells?

- As process nodes shrink and we move from planar transistors to 22FDX, the number of DRC rules has increased exponentially. This brings large number of challenges and constraints from layout of Standard Cell to full chip place and route. One area of increased complexity that affects place and route is the periphery of the logic block.
- Finishing cells/ENDCAP cells are used to ensure the periphery of logic blocks to meet all DRC rules. For 28 nm and larger process, there is only one type of finishing cell needed on left and right. But in advanced technologies, in addition to left and right; top, bottom, concave and convex corners are also introduced.

7.1.2 Different Corners and Edges

Figure 7-1 Different Corners and Edges - 1

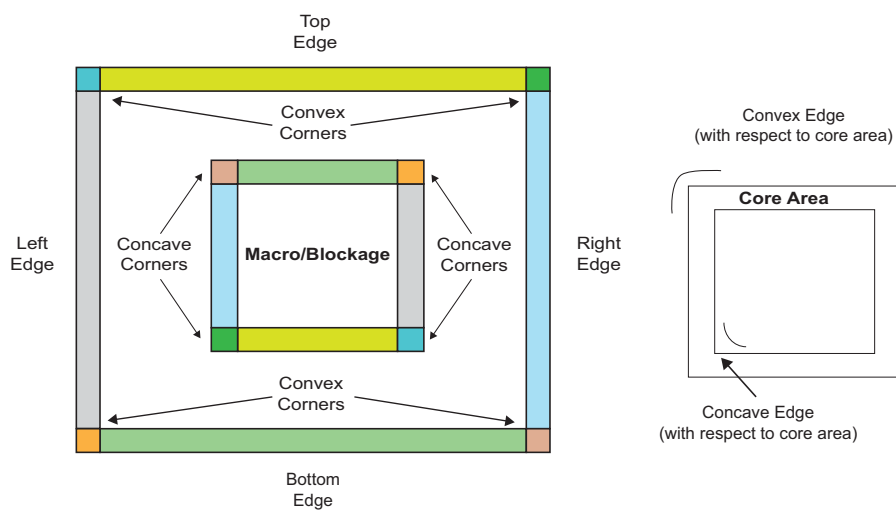
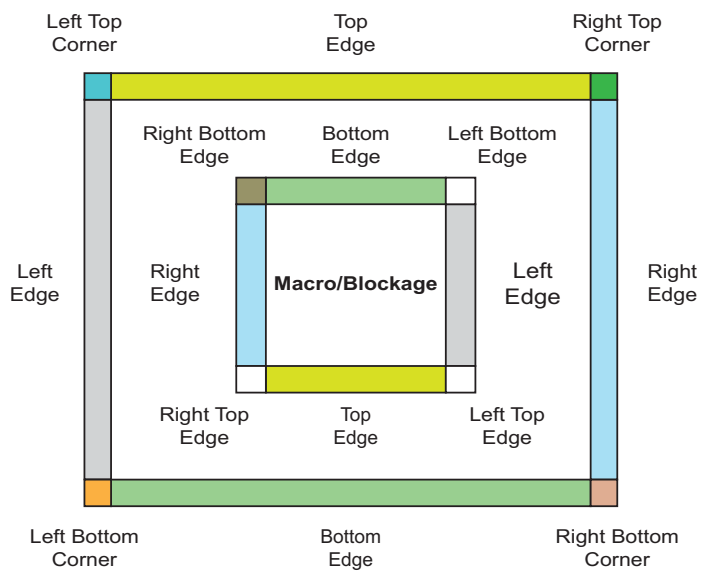
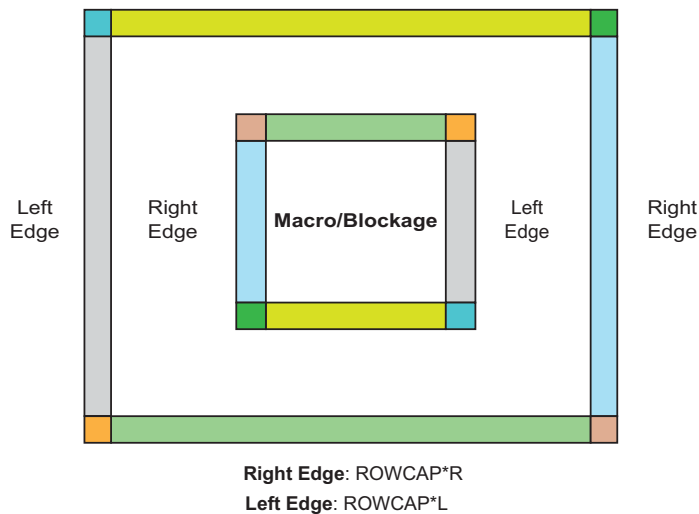


Figure 7-2 Different Corners and Edges - 2



7.1.3 ROWCAP Cells

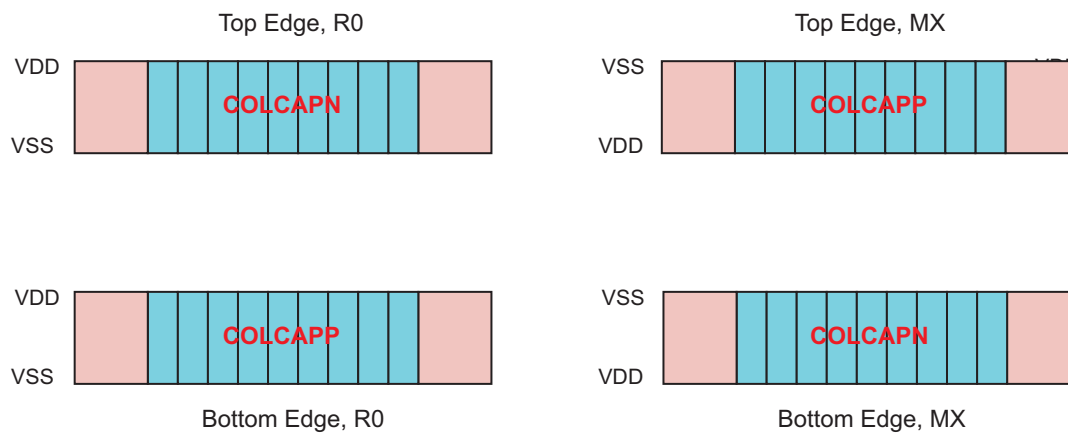
Figure 7-3 ROWCAP Cells



7.1.4 COLCAP Cells Placement

Place proper COLCAPN and COLCAPP cells based on the row orientation and column edge.

Figure 7-4 COLCAP Cells Placement



7.1.5 Convex Corner Cell Placement

Place proper convex corner cell based on the corner and row orientation.

Figure 7-5 Convex Corner Cell Placement

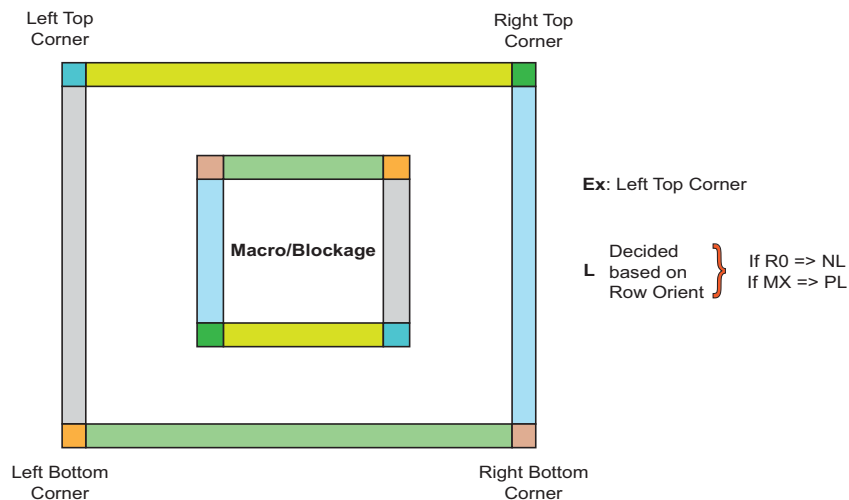


Table 7-1 Convex Corner Cell Placement

Placement	Row Orientation	Cell Type
Left Top Corner	R0	CNREXTANTENNANL
	MX	CNREXTANTENNAPL
Right Top Corner	R0	CNREXTANTENNANR
	MX	CNREXTANTENNAPR
Left Bottom Corner	R0	CNREXTANTENNAPL
	MX	CNREXTANTENNANL
Right Bottom Corner	R0	CNREXTANTENNAPR
	MX	CNREXTANTENNANR

7.1.6 Concave Corner Cell Placement

Place proper concave cells based on the corner and row orientation.

Figure 7-6 Concave Corner Cell Placement

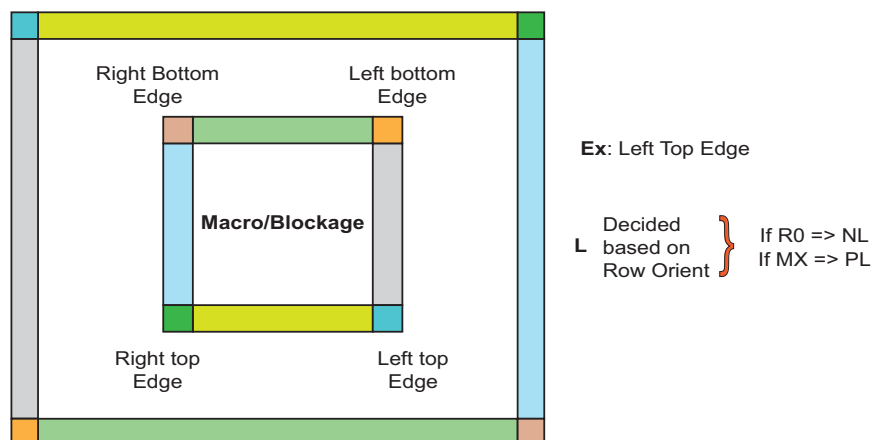


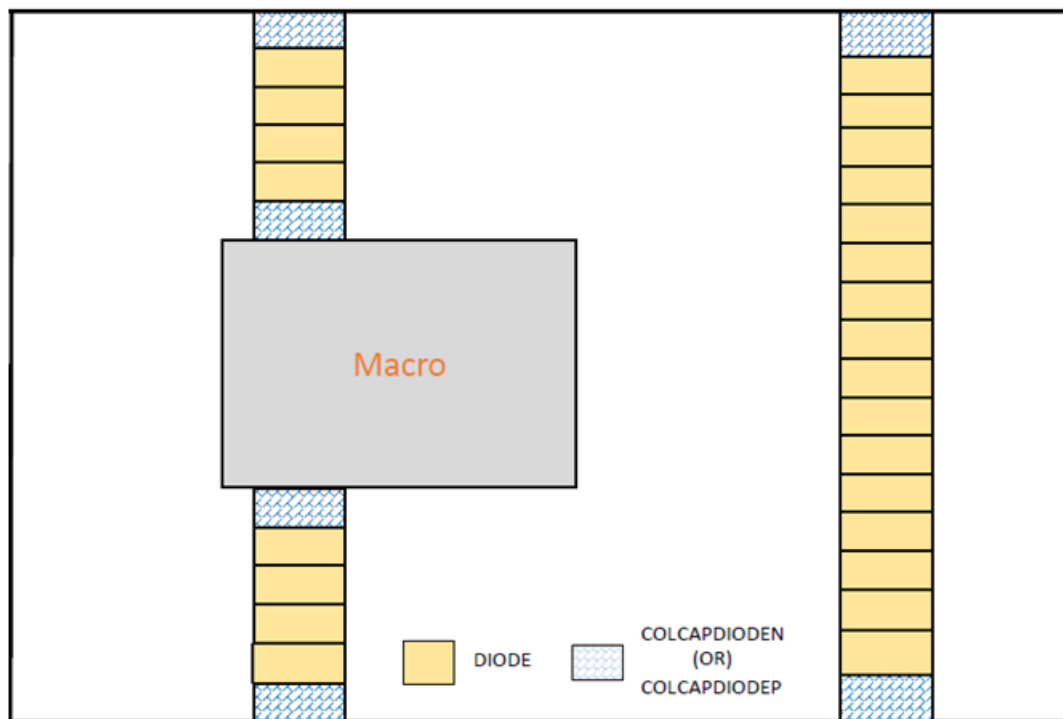
Table 7-2 Concave Corner Cell Placement

Placement	Row Orientation	Cell Type
Left Top Corner	R0	CONCAVENL
	MX	CONCAVEPL
Right Top Corner	R0	CONCAVENR
	MX	CONCAVEPR
Left Bottom Corner	R0	CONCAVEPL
	MX	CONCAVENL
Right Bottom Corner	R0	CONCAVEPR
	MX	CONCAVENR

7.1.7 Core Diode Cells Placement

The core diode cells will terminate at the block edges with special termination cells based on the row orientation.

Figure 7-7 Core Diode Cells



The core diode cells will terminate at the block edges with special termination cells based on the row orientation as shown in the following table.

Table 7-3 Core Diode Cell Placement

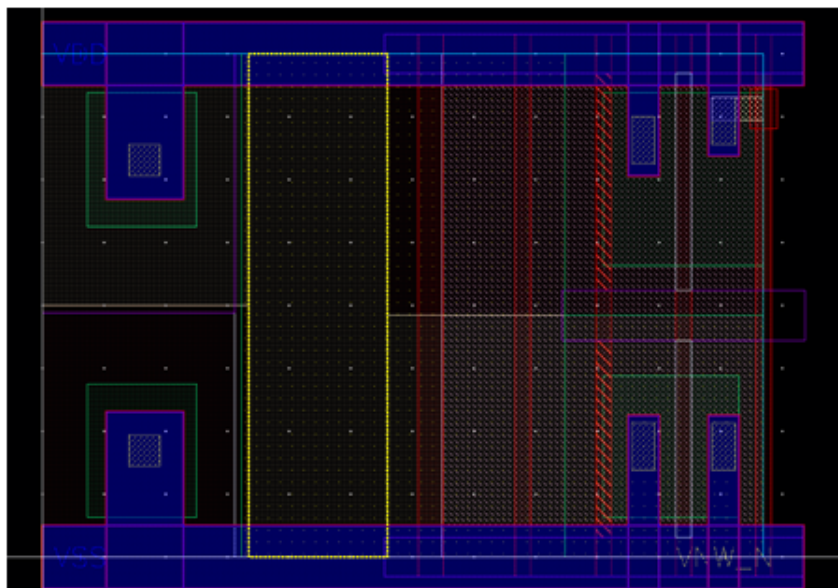
Location	Row Orientation	Cell Type
Top Tap Cell	RO	COLCAPDIODEN
	MX	COLCAPDIODEP
Bottom Tap Cell	RO	COLCAPDIODEP
	MX	COLCAPDIODEN

7.2 Details of Physical Cell Guidelines

7.2.1 ROWCAPANTENNAL

- **Function:** It avoids the enclosure DRCs of layers on Left side. It also provides the discharging path for antenna of VDD rail to p-substrate through DIODE and connects VSS rail to p-substrate directly.
- **Usage:** Place the cell at left side of the block.

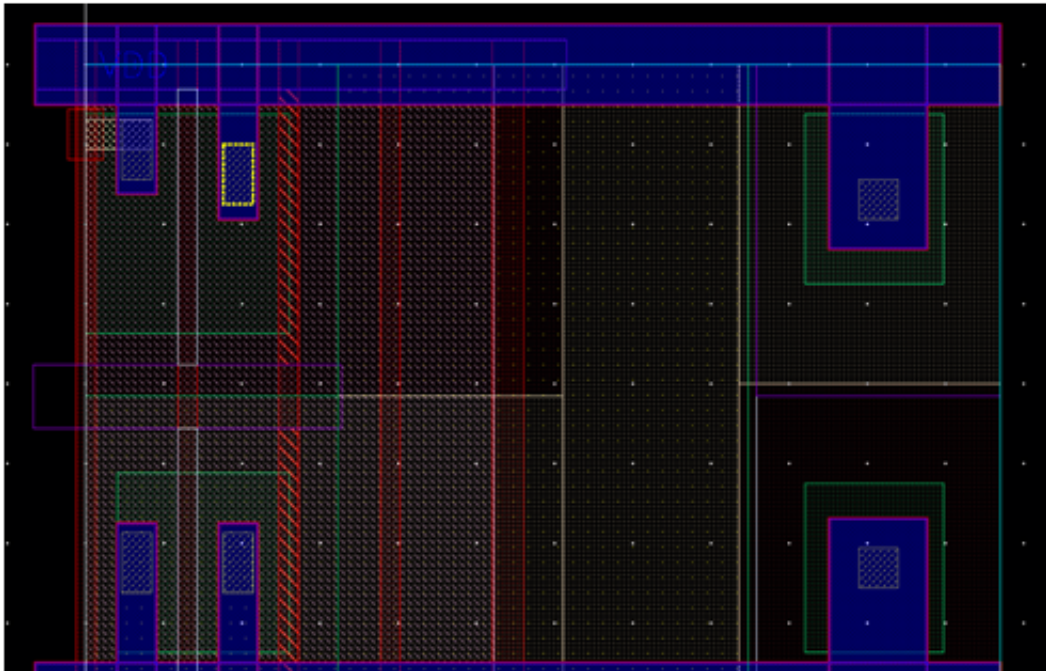
Table 7-4 ROWCAPANTENNAL



7.2.2 ROWCAPANTENNAR

- **Function:** It avoids the enclosure DRCs of layers on right side. It also provides the discharging path for antenna of VDD rail to p-substrate through DIODE and connects VSS rail to p-substrate directly.
- **Usage:** Place the cell at right side of the block.

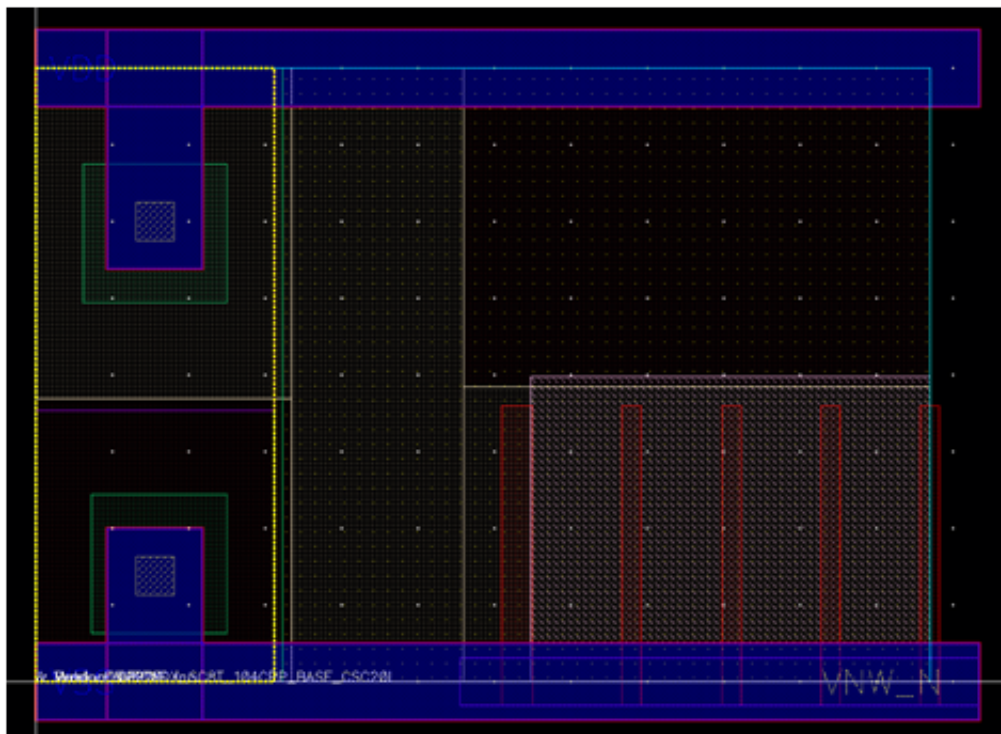
Figure 7-8 ROWCAPANTENNAR



7.2.3 CNREXTANTENNANL

- **Function:** It avoids the enclosure DRCs of layers at NWELL left corner location. It also provides the substrate connection to VSS which in turn avoids the VSS rail antenna violation.
- **Usage:** Place the cell at NWELL left corner location of the block.

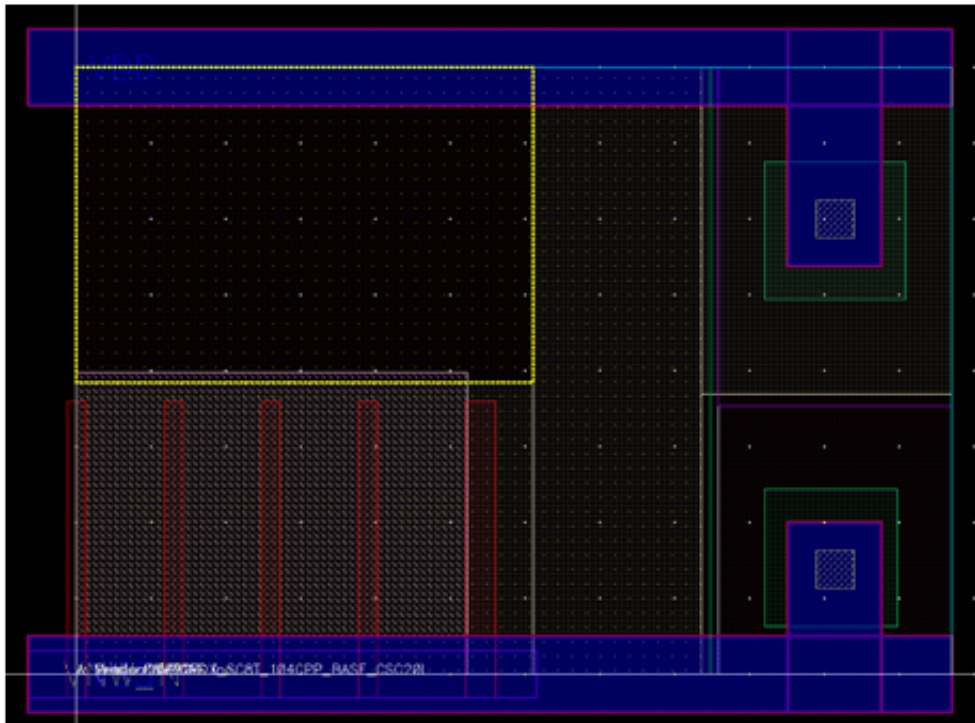
Figure 7-9 CNREXTANTENNANL



7.2.4 CNREXTANTENNANR

- **Function:** It avoids the enclosure DRCs of layers at NWELL right corner location. It also provides the substrate connection to VSS which in turn avoids the VSS rail antenna violation.
- **Usage:** Place the cell at NWELL right corner location of the block.

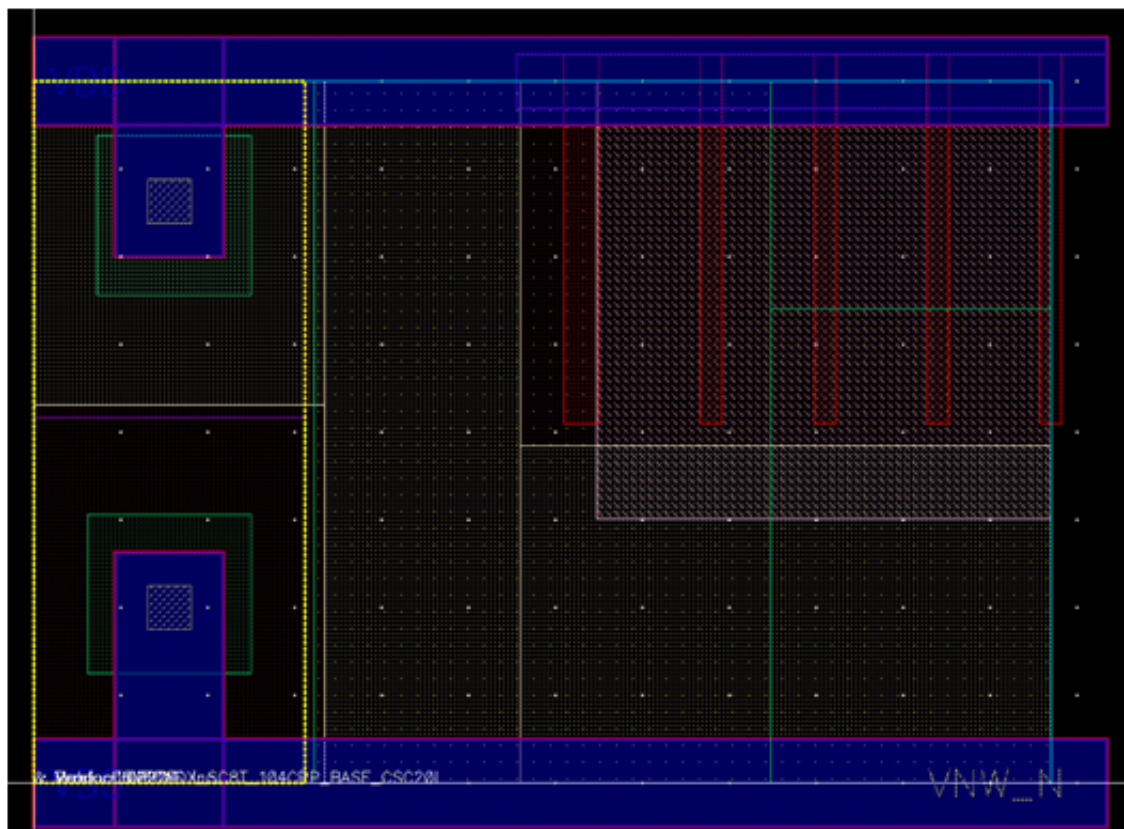
Figure 7-10 CNREXTANTENNANR



7.2.5 CNREXTANTENNAPL

- **Function:** It avoids the enclosure DRCs of layers at PWELL left corner location. It also provides the discharging path for antenna of VDD rail to p-substrate through DIODE.
- **Usage:** Place the cell at PWELL left corner location of the block.

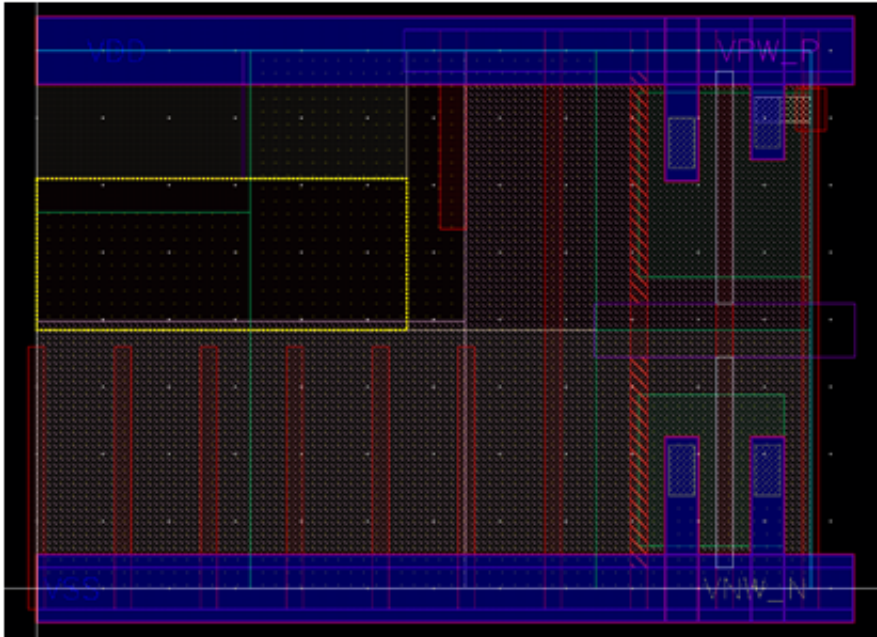
Figure 7-11 CNREXTANTENNAPL



7.2.6 CNREXTANTENNAPR

- **Function:** It avoids the enclosure DRCs of layers at PWELL right corner location. It also provides the discharging path for antenna of VDD rail to p-substrate through DIODE.
- **Usage:** Place the cell at PWELL right corner location of the block.

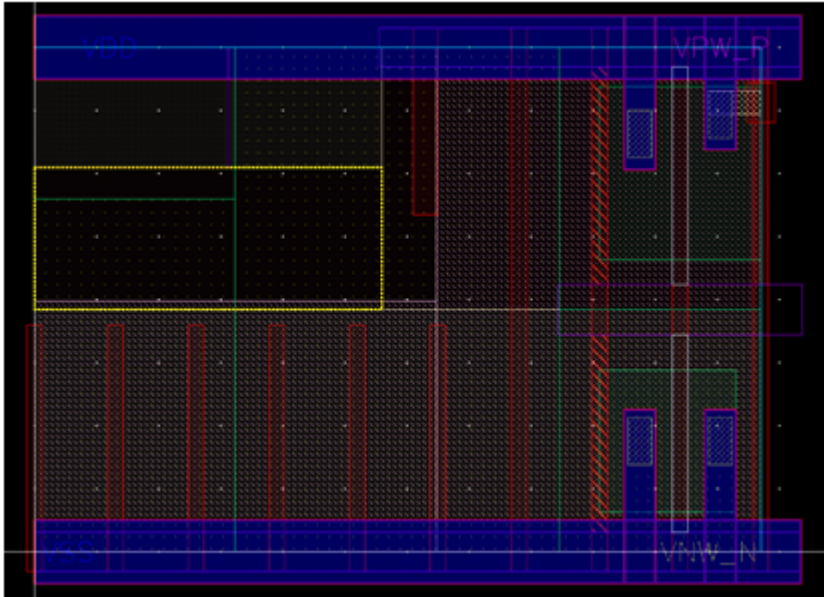
Figure 7-12 CNREXTANTENNAPR



7.2.7 CONCAVENL

- **Function:** It avoids the enclosure DRCs of layers at N-WELL left side location and provides the layers continuity for PWELL side.
- **Usage:** Place the cell at left concave edge of the block which ends with N-WELL.

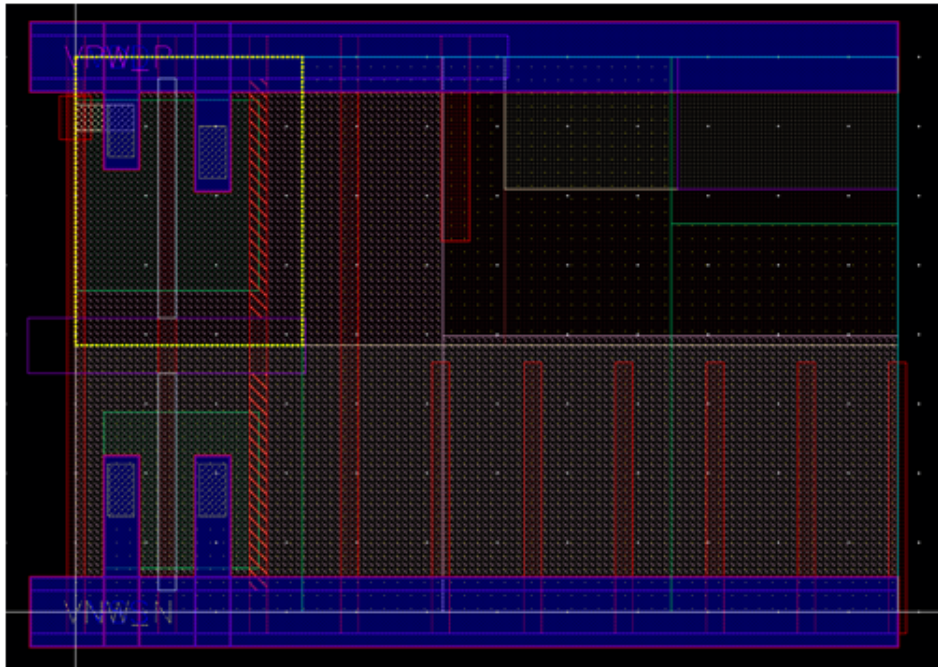
Figure 7-13 CONCAVENL



7.2.8 CONCAVENR

- **Function:** It avoids the enclosure DRCs of layers at N-WELL right side location and provides the layers continuity for P WELL side.
- **Usage:** Place the cell at right concave edge of the block which ends with N-WELL.

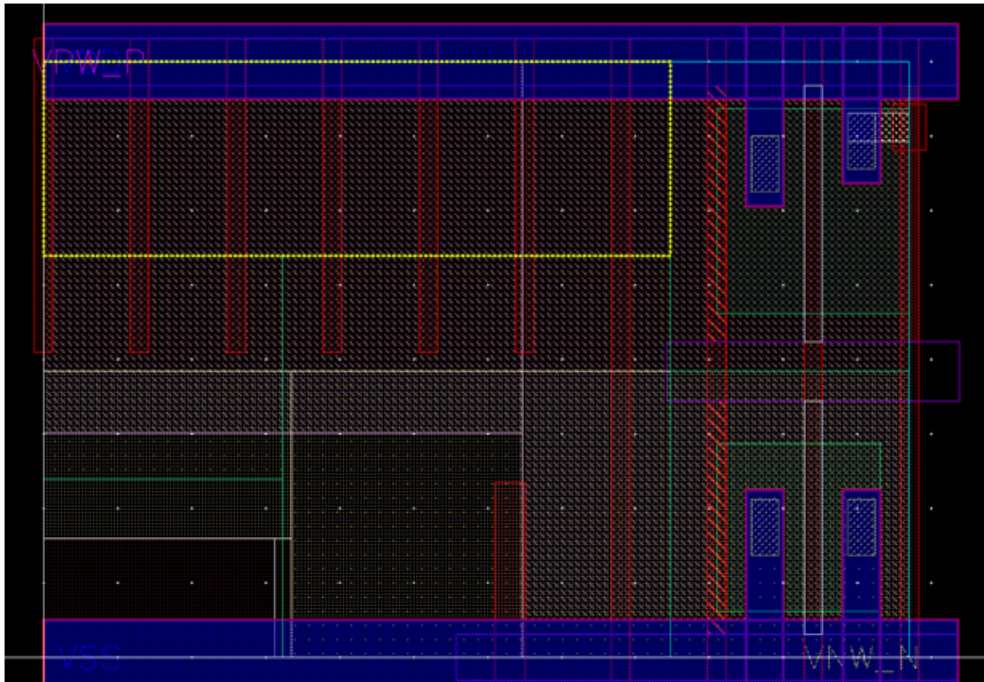
Figure 7-14 CONCAVENR



7.2.9 CONCAVEPL

- **Function:** It avoids the enclosure DRCs of layers at P-WELL left side location and provides the layers continuity for NWELL side.
- **Usage:** Place the cell at left concave edge of the block which ends with P-WELL.

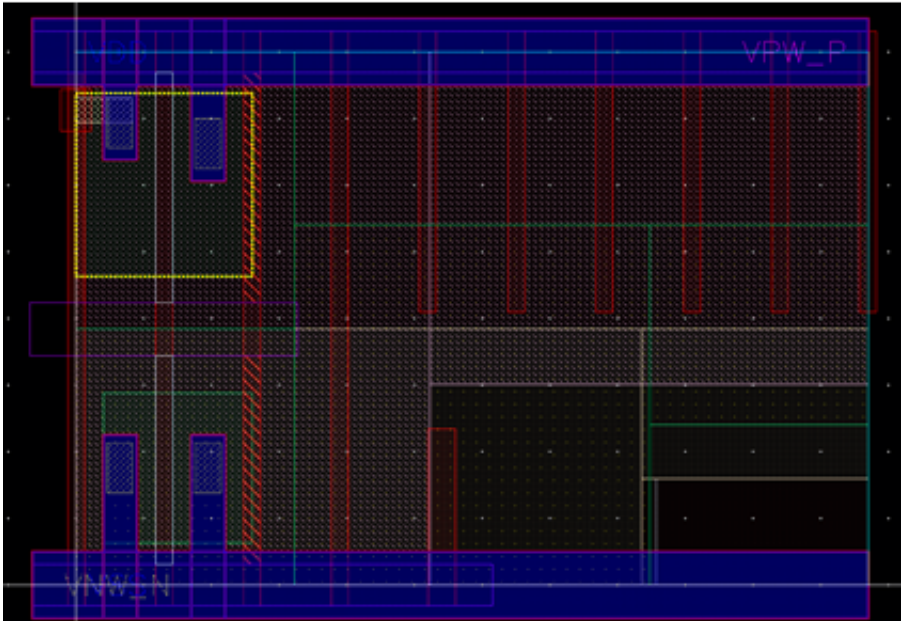
Figure 7-15 CONCAVEPL



7.2.10 CONCAVEPR

- **Function:** It avoids the enclosure DRCs of layers at P-WELL right side location and provides the layers continuity for N-WELL side.
- **Usage:** Place the cell at right concave edge of the block which ends with P-WELL.

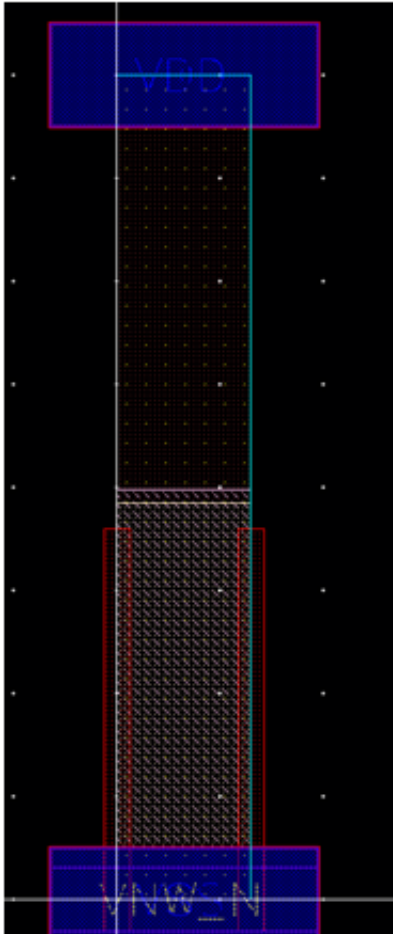
Figure 7-16 CONCAVEPR



7.2.11 COLCAPN

- **Function:** It avoids the enclosure DRCs of layers at N-WELL top/bottom location.
- **Usage:** Place the cell at top/bottom location of the block which ends with N-WELL.

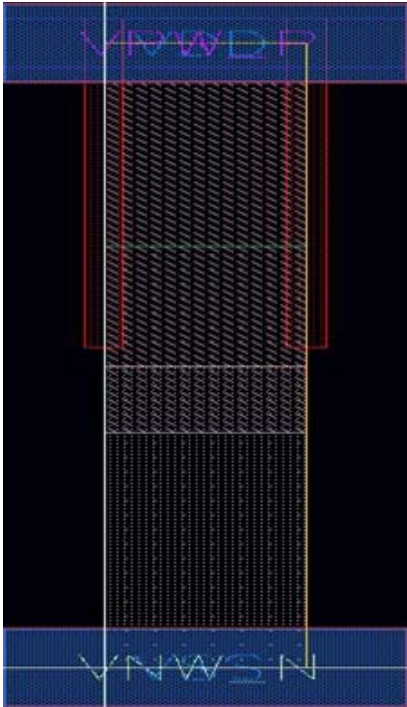
Figure 7-17 COLCAPN



7.2.12 COLCAPP

- **Function:** It avoids the enclosure DRCs of layers at P-WELL top/bottom location.
- **Usage:** Place the cell at top/ bottom location of the block which ends with P-WELL.

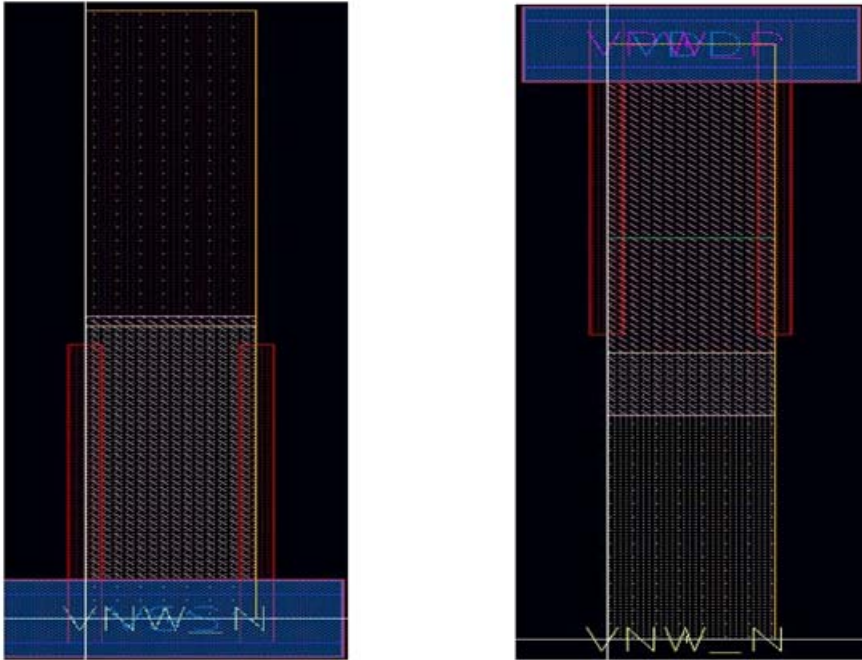
Figure 7-18 COLCAPP



7.2.13 NOM1COLCAPN and NOM1COLCAPP

- **Function:** Same as COLCAP cells by removing M1 on VDD (in N-Type)/VSS (in P-Type). This may be useful to avoid antenna violation on WELL for longer blocks.

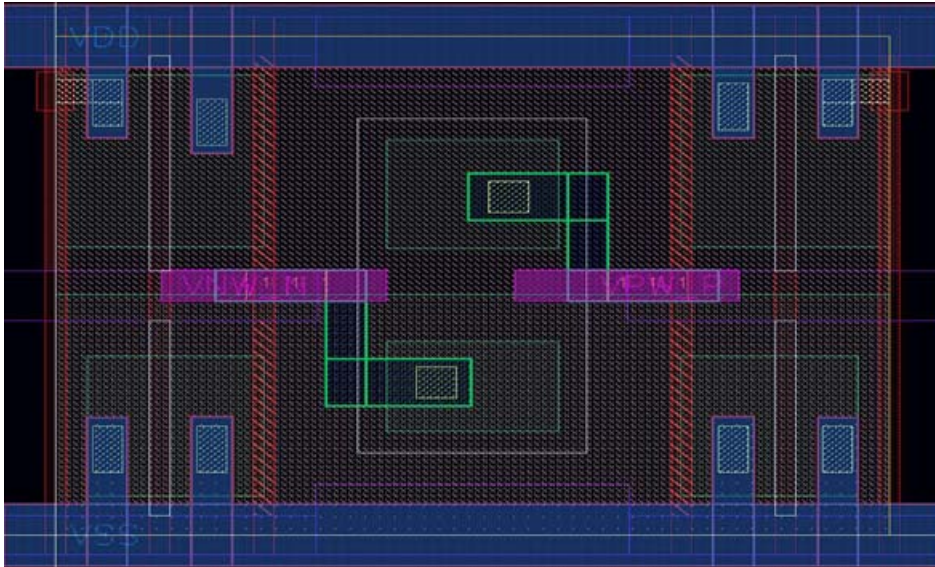
Figure 7-19 NOM1COLCAPN and NOM1COLCAPP



7.2.14 TAP

- **Function:** Connects PMOS and NMOS bulk terminals to respective bias signals.
- **Usage:** Place the cell at regular intervals of design. Route PMOS bulk terminal VPW_P and NMOS bulk terminal VNW_N to required bias nets.

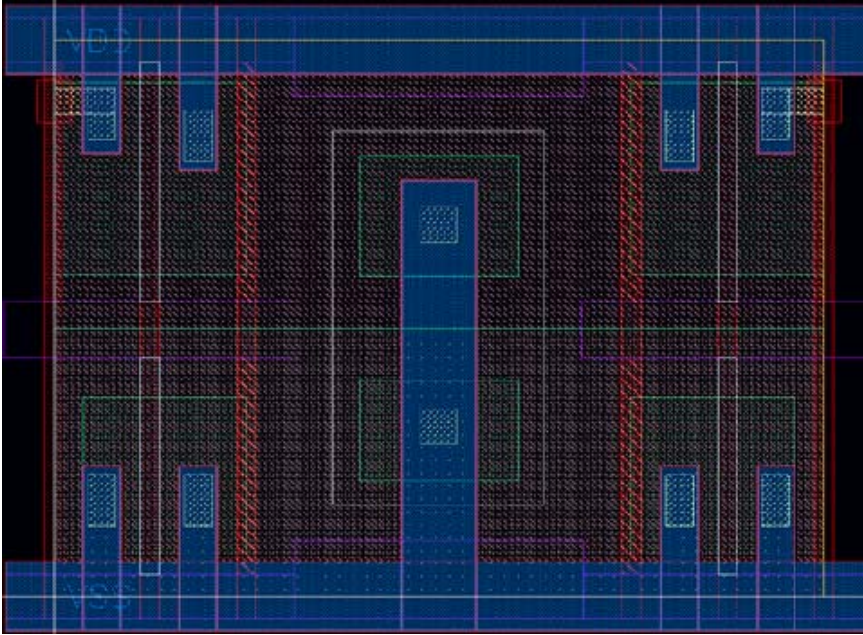
Figure 7-20 TAP



7.2.15 TAPZB

- **Function:** Connects the PMOS and NMOS bulk terminals to VSS.
- **Usage:** Place the cell at regular intervals of design.

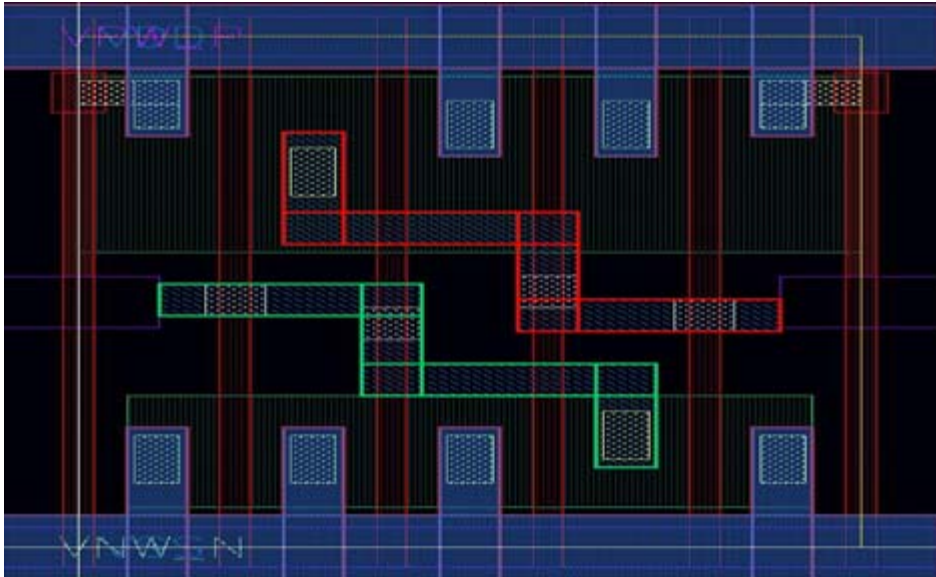
Figure 7-21 TAPZB



7.2.16 DECAP

- **Function:** It provides decoupling between VDD and VSS rails
- **Usage:** DECAP cells insertion before fill is recommended

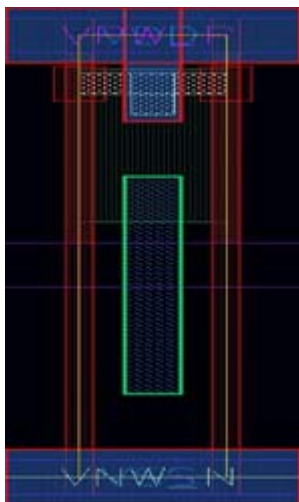
Figure 7-22 DECAP



7.2.17 1CPP FILL

- Cadence and Synopsys based flows support 1CPP aware placement, but make sure to use FILLX1 cell, i.e should support 1CPP aware placement only between DRIAN-DRAIN Combination. 1CPP placement gap should be prohibited in rest of the cases.

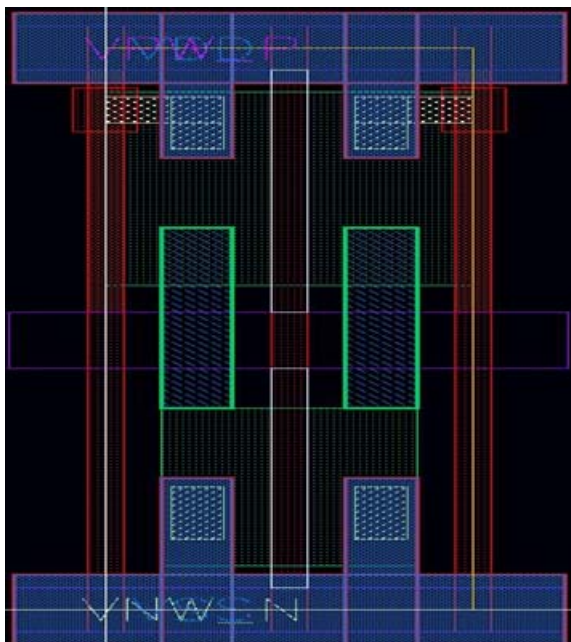
Figure 7-23 CPP FILL



7.2.18 FILL

- **Function:** Filler cells need to be placed in empty areas for WELL and implant layers continuity.
- **Usage:** To fill the empty area.

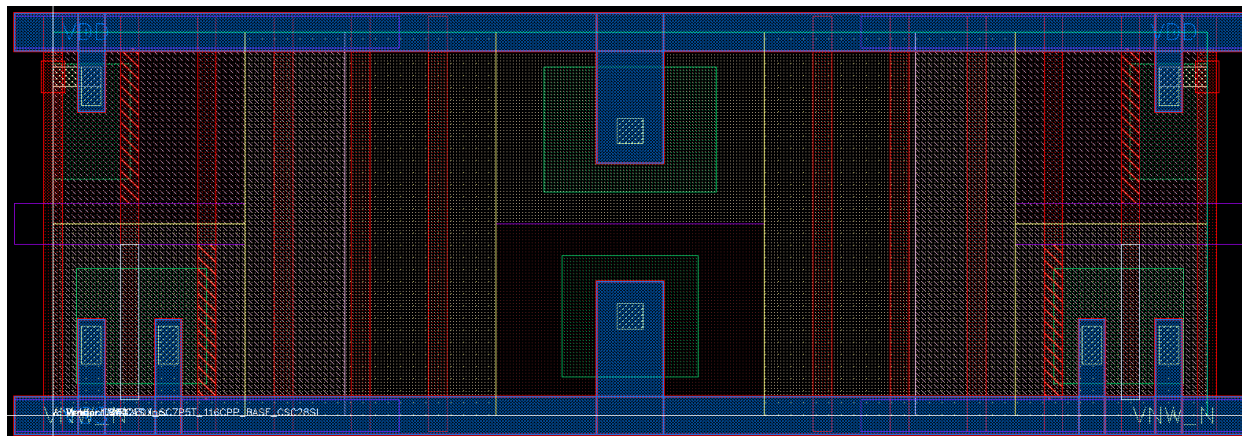
Figure 7-24 FILL



7.2.19 Core Diode Cells

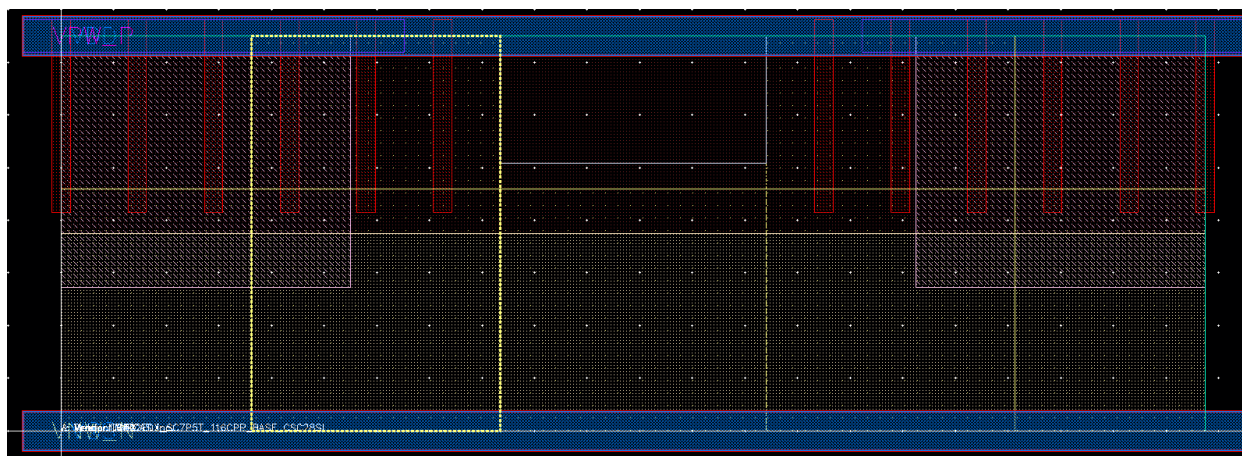
7.2.19.1 CORE DIODE CELL

Figure 7-25 CORE DIODE CELL



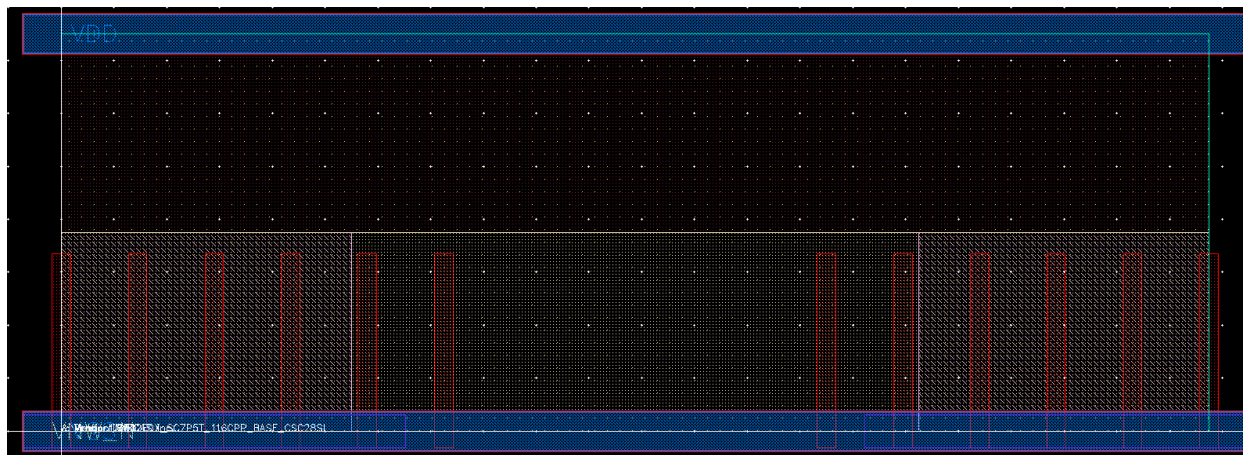
7.2.19.2 COLCAPDIODEP CELL

Figure 7-26 COLCAPDIODEP CELL



7.2.19.3 COLCAPDIODEN CELL

Figure 7-27 COLCAPDIODEN CELL

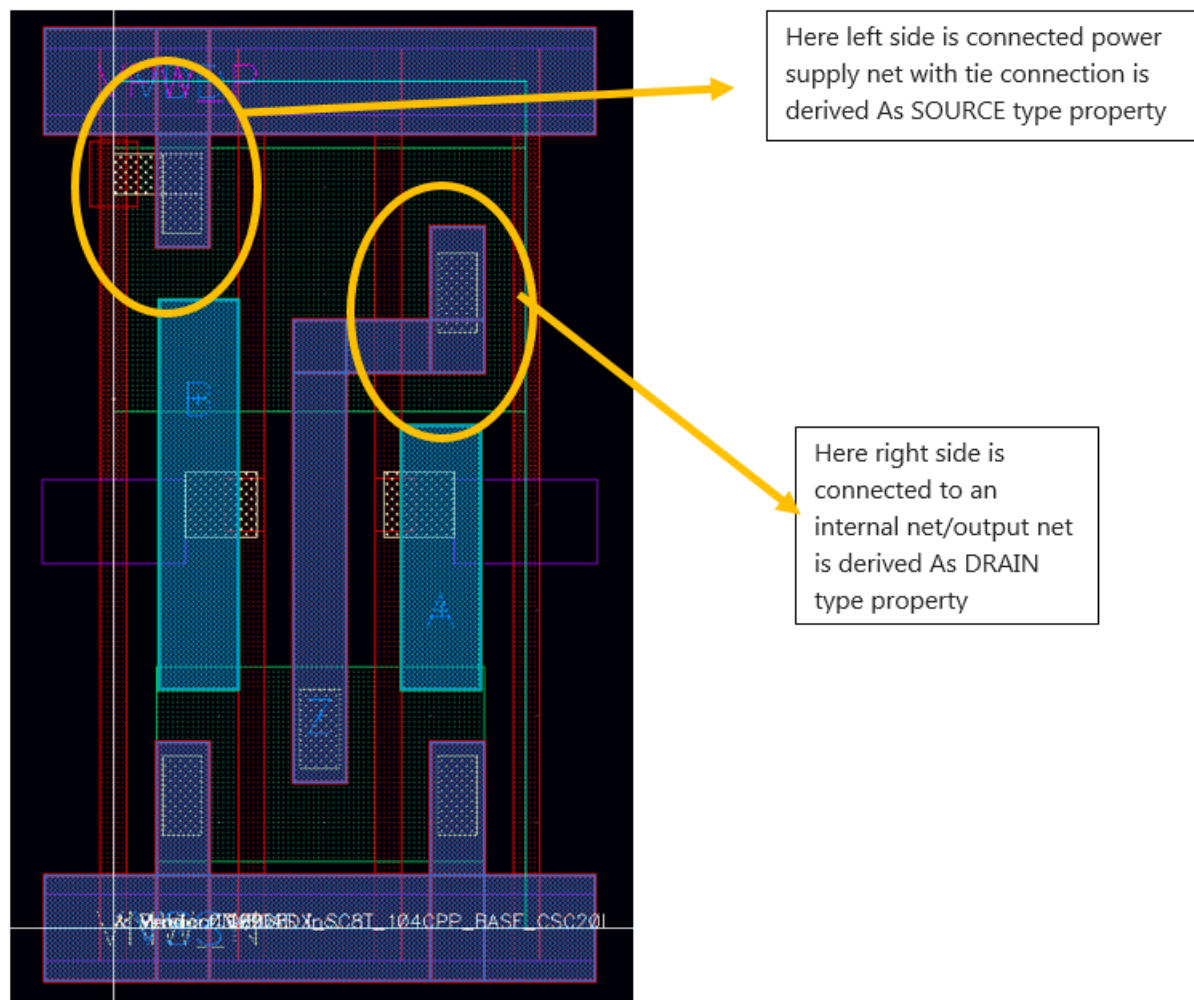


Follow the below guidelines for core diode cell placement:

1. Placement of core diode cells should be in a vertical stack pattern (No checkered pattern) to avoid getting any DRC violations.
2. Between any two core diode cells, vertical abutment should be exact and no offset is to be present.
3. These cells should be terminated at the core boundaries i.e top and bottom rows with their corresponding *COLCAPDIODE* cells.
4. Tap placement of cells should be mandatory in between the core diode cells, i.e when these cells are separated horizontally and ROW termination cells (*ROWCAP*) and core diode cells

7.2.20 Motivation for Placement Guidelines

- 22FDX 8T libraries require additional placement constraints for controlling Standard Cell abutment or spacing between Standard Cells.
- The need for placement rules are driven by DFM requirements and cell design (PMOS is CNRX - Continuous RX).
- The correct usage of the placement constraints are verified during DRC.
- Each cell contains two types of edge constraints, which is SOURCE or DRAIN property.
- Edge properties are derived based on cell left/right edge PMOS connection.

Figure 7-28 Placement Guidelines

We can about the combination specified in the following table for all the cells.

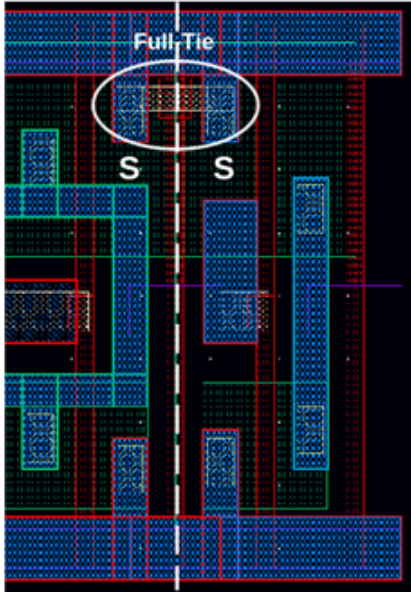
Table 7-5 Placement Constraints

8 T CNRX	Source-Source	Source-Drain	Source2CPP-Source2CPP	Source2CPP-Source	Source2CPP-Drain	Drain-Drain
Abutment	Yes	Yes	Yes	Yes	No	No
1x Spacing	No	No	No	No	No	Yes
2 x Spacing	Yes	Yes	Yes	Yes	Yes	Yes

7.2.20.1 Source-Source

Non-Failure Case: The following figure shows the formation of a U-construct when two adjacent source type cells are abutting, which prevents the occurrence of double attacker.

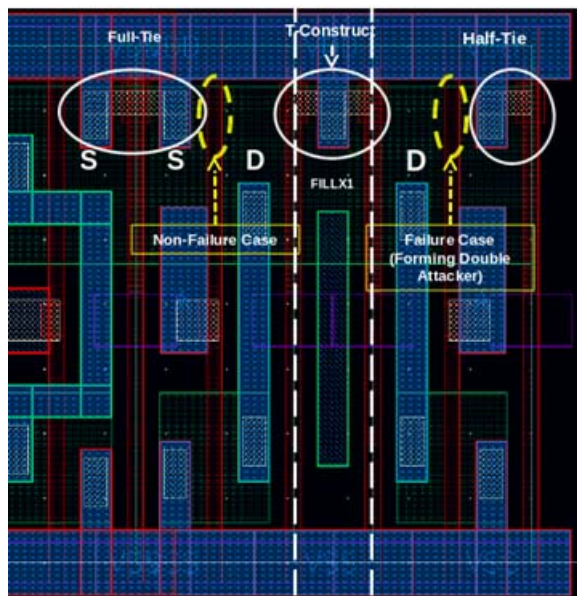
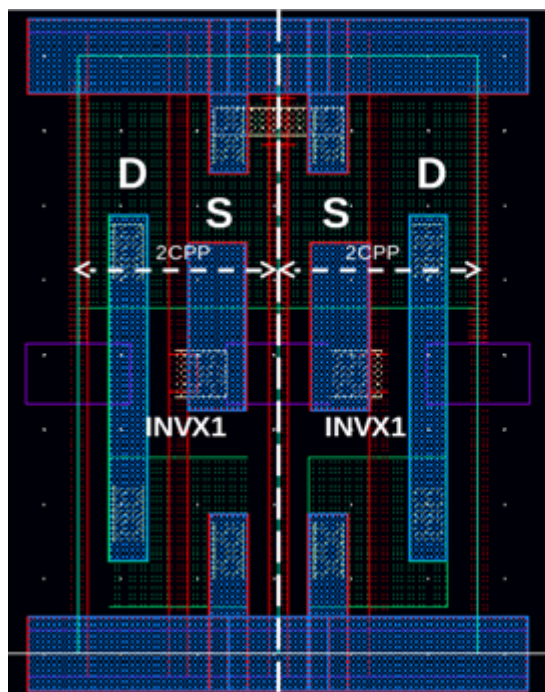
Figure 7-29 Source - Source



7.2.20.2 Double Attacker Formation

Failure Case: The active gate which is present between T-construct and half-tie will prone the double attacker which is shown in the right side of the following figure.

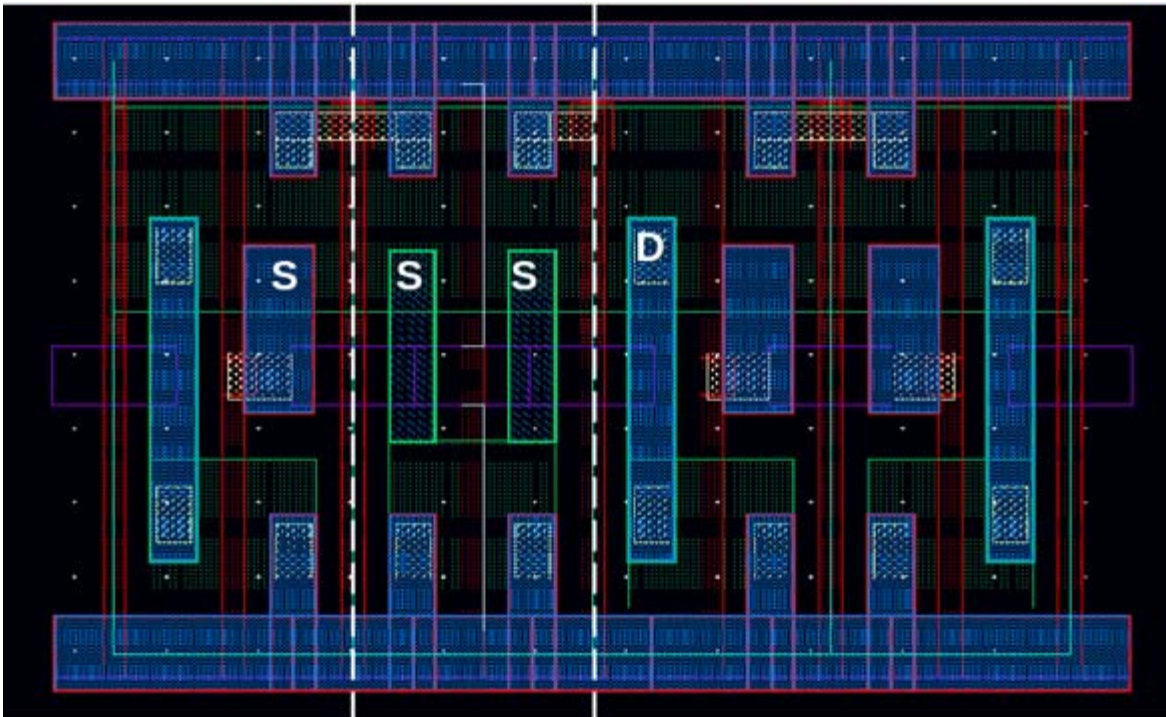
Non-Failure Case: Even the active gate is present between constructs and double attacker will not occur due to formation of U-construct, highlighted left side of the following figure.

Figure 7-30 Double Attacker Formation**7.2.20.3 Source2CPP - Source2CPP****Figure 7-31 Source2CPP - Source2CPP**

S = SOURCE
D = DRAIN

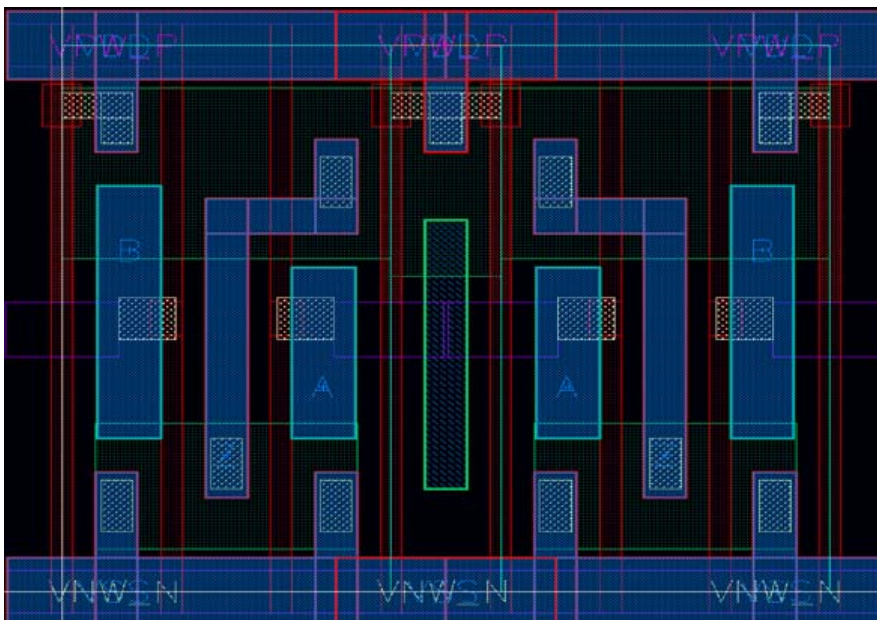
7.2.20.4 Source2CPP/Source-Drain Fill X2

Figure 7-32 Source2CPP/Source-Drain Fill X2



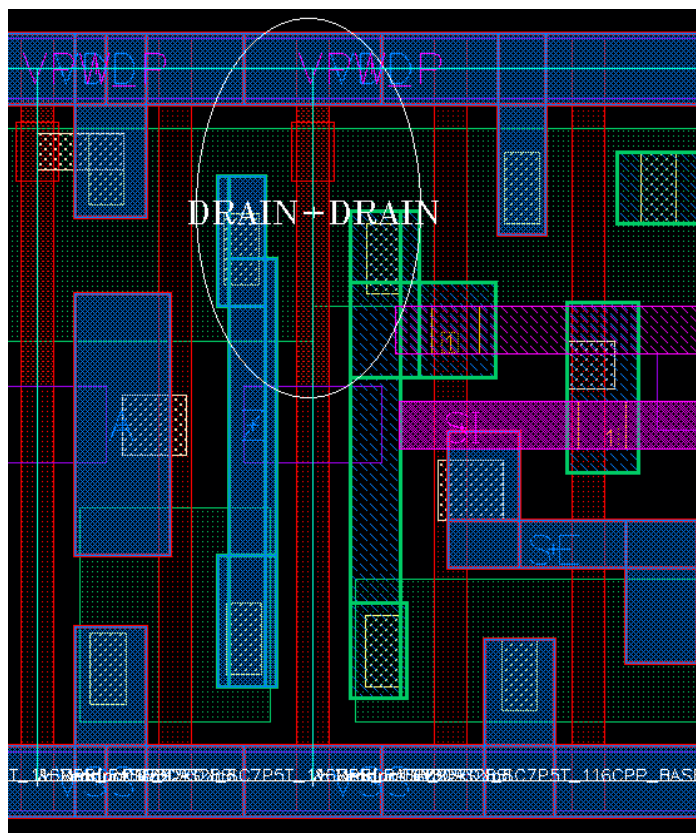
7.2.20.5 Drain-Drain Combination

Figure 7-33 Drain-Drain Combination



Case 1: Drain-Drain Combination

Figure 7-34 CASE 1: Drain-Drain Combination

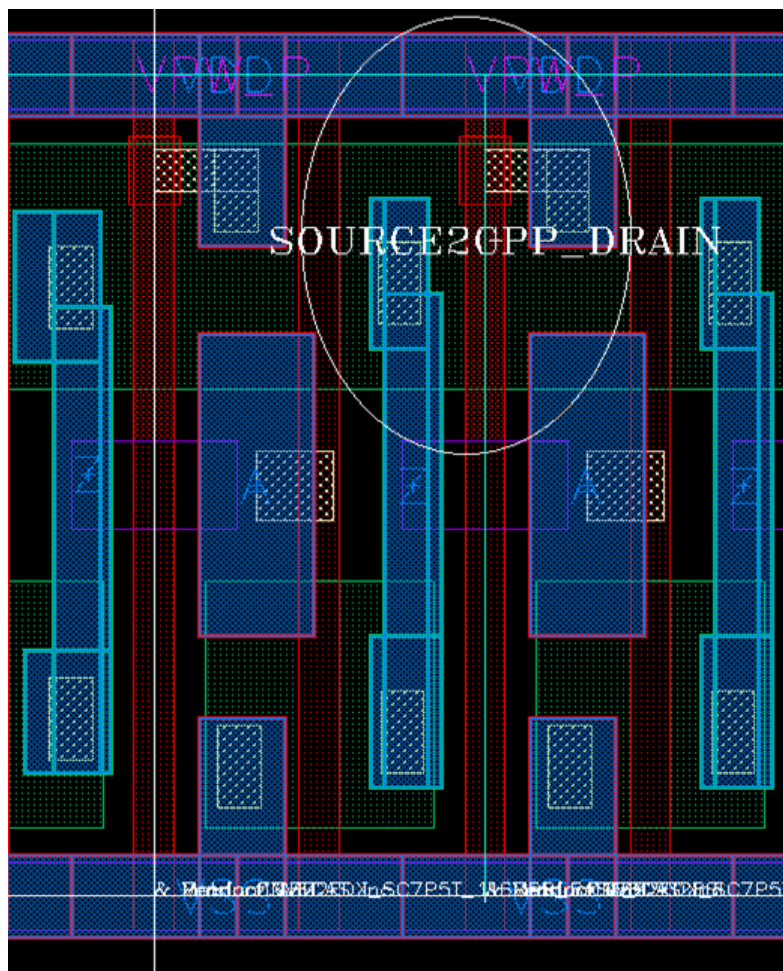
**GRAUX3.C.9**

AUXPC3 must incur Union (PC_Tab, (((PC not CT) with width == 0.036 μm) touching Union (CnRX_CB_Half_ULP, CnRX_CB_Full_ULP, CnRX_CB_Filler_ULP)))

```
(PC_Tab C-edge not touching Union (CnRX_CB_Filler, CnRX_CB_OneAndHalfFiller,
CnRX_CB_Filler_ULP, CnRX_CB_Half_ULP, CnRX_CB_2CPPFiller)) with space to (FLTGate, AUXPC2) >
0.076 μm) minimum
C- space >= 0.276
```

Case 2: Source2CPP-Drain Combination

Figure 7-35 Source2CPP-Drain Combination



7.2.20.8 DRC Seen

GRCnRXULP.PC.S.2

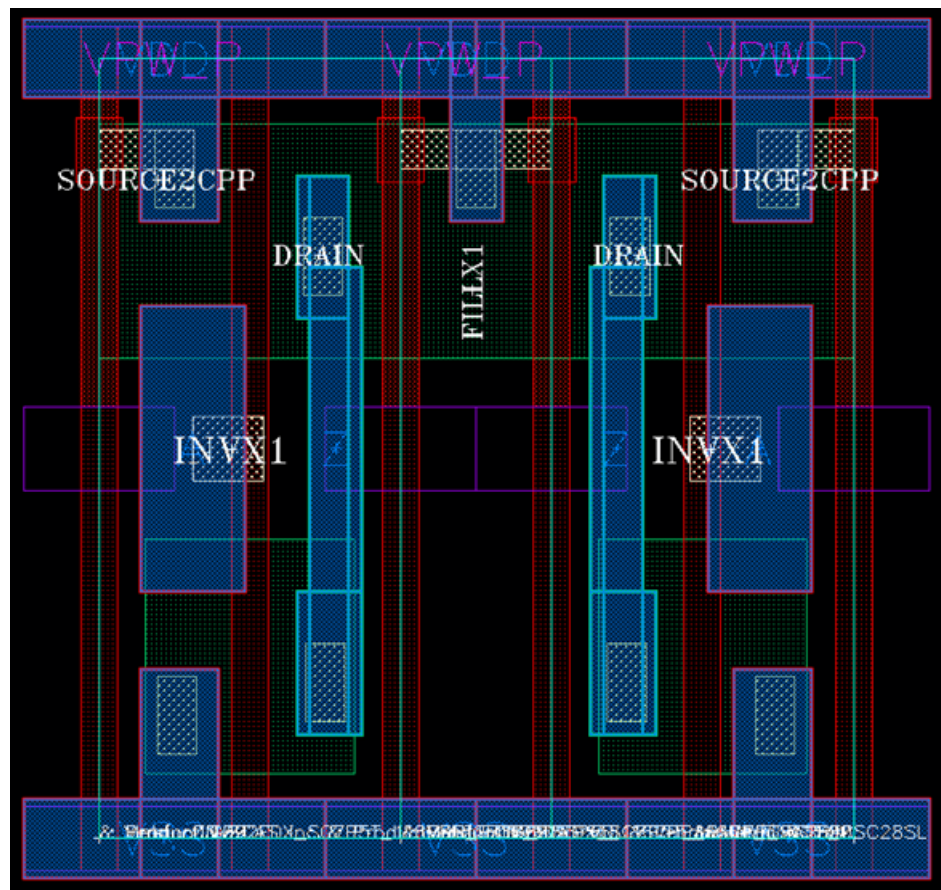
((PC_Tab touching Union (ULPN, ULPP)) C-edge not touching CnRX_CB_Filler_ULP) with space to (FLTGATE, AUXPC2) > 0.084 μm) minimum C-space >= 0.312

GRCnRXULP.C.5

PC_Tab touching Union (ULPN, ULPP) must incur Union (CnRX_CB_Half_ULP, CnRX_CB_Full_ULP, CnRX_CB_Filler_ULP).

Case 3: Drain_Drain_Drain Combination

Figure 7-36 Drain-Drain-Drain Combination



7.2.20.9 DRC Seen

GRCnRXULP.PC.S.2

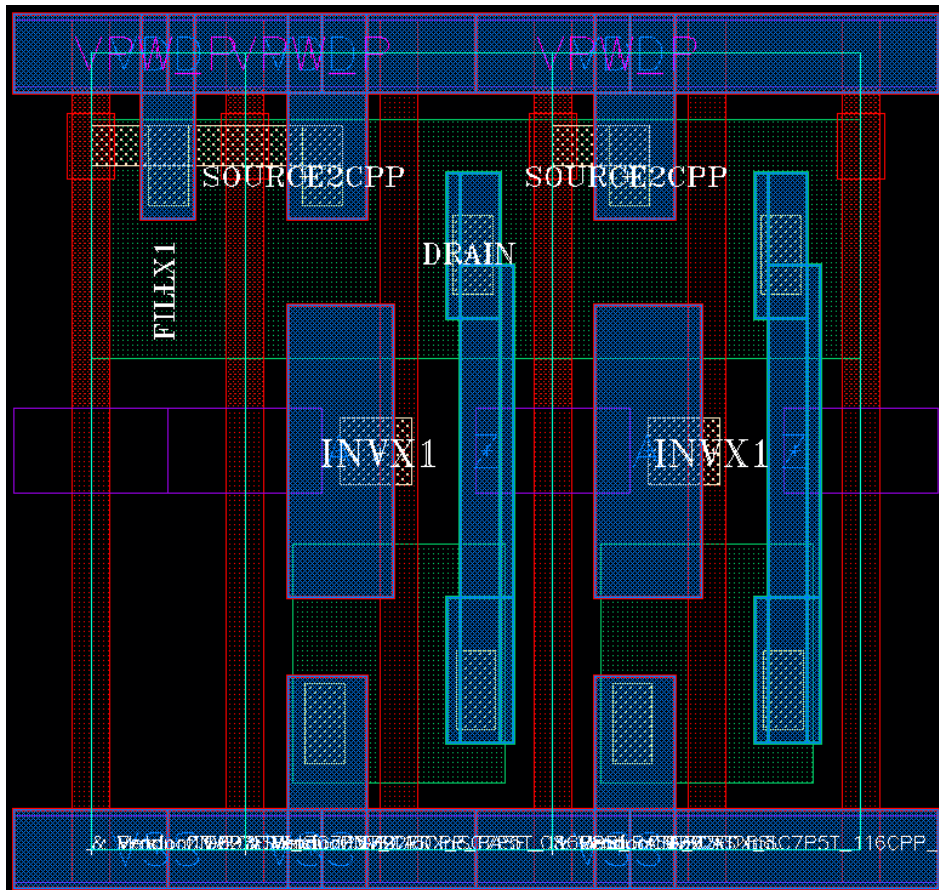
((PC_Tab touching Union (ULPN, ULPP)) C-edge not touching CnRX_CB_Filler_ULP) with space to (FLTGATE, AUXPC2) > 0.084 μm) minimum C-space >= 0.312

GRCnRXULP.C.5

PC_Tab touching Union (ULPN, ULPP) must incur Union (CnRX_CB_Half_ULP, CnRX_CB_Full_ULP, CnRX_CB_Filler_ULP).

Case4: Source2CPP-Drain-Source2CPP

Figure 7-37 Source2CPP-Drain-Source2CPP



7.2.20.10 DRC Seen

GRCnRXULP.PC.S.2

((PC_Tab touching Union (ULPN, ULPP)) C-edge not touching CnRX_CB_Filler_ULP) with space to (FLTGATE, AUXPC2) > 0.084 μm) minimum C-space >= 0.312

GRCnRXULP.C.5

PC_Tab touching Union (ULPN, ULPP) must incur Union (CnRX_CB_Half_ULP, CnRX_CB_Full_ULP, CnRX_CB_Filler_ULP).

